Critical Design Report
Lifting Large Design Co.

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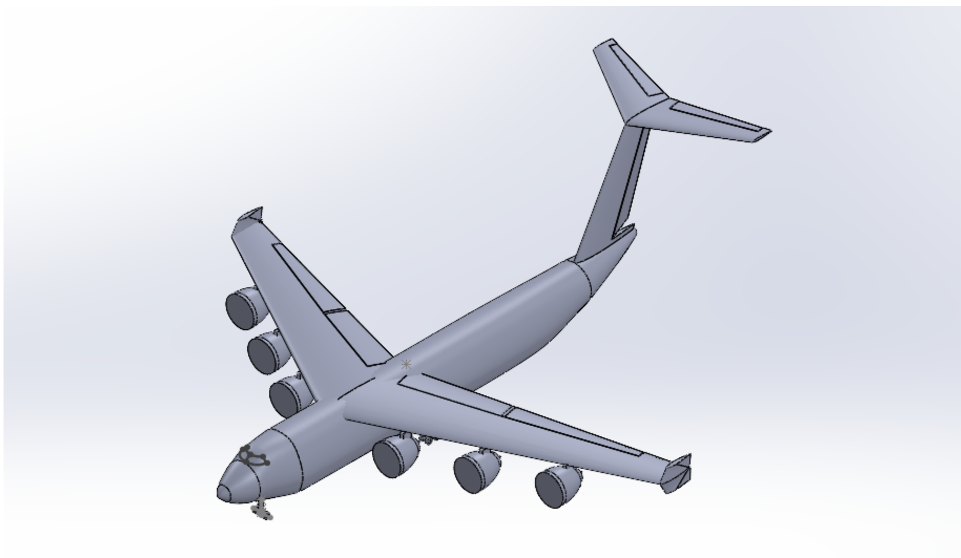

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Executive Summary (Ian)

Approached by the United States Department of Defense (DoD), Lifting Large has been asked to complete a heavy-lift aircraft design that will be used as a replacement for the aging United States Air Force's (USAF) currently in service C-5 and C-17 cargo transports with an entry into service (EIS) date of 2033. Outside of an increased max payload requirement of 430,000 pounds (lbs), the replacement aircraft must be able to deliver the max payload at a range of 2,500 nautical miles (nmi). A reduced payload of 295,00 lbs must be carried over 5,000 nmi, and a no payload (ferry) configuration must be capable of traveling 8,000 nmi. Along with meeting the base requirements, the aircraft must abide by INternational Civil Aviation Organization (ICAO) Class F airport regulations and Federal Aviation Regulations (FAR) part 25. This report details the final design, design process, analysis, validation, and finally, the cost of the aircraft developed by Lifting Large.

Given the requirements, Lifting Large determined a conventional tube and wing design would best suit the needs of the stakeholder. Noting the extremely large payload size and profit margin requirements, Lifting Large targeted reducing the aircraft empty weight as much as possible. Compared to other possible configurations, such as a blended wing body (BWB) or braced wing, a tube and wing design simplifies the structure of the aircraft at the slight cost of a reduction in the aerodynamic efficiency as opposed to a BWB. In addition, less support structure is needed for the aircraft features compared to a BWB or braced wing which drives down the cost, allowing for greater profit margin. Finally, as a tube and wing design has been in use much longer than the two other options, Lifting Large felt more confident in the aircraft design succeeding as a tube and wing given the extensive testing and publicly available information.

Moving further into the design choices, Lifting Large determined a high wing, t-tail configuration to best meet both the requirements, and the stakeholder needs. As noted by the USAF, the aircraft must be capable of efficient loading using minimal ground equipment. With this in mind, both the high wing and t-tail design choices aid in optimizing the loading/unloading procedure. Due to the high ground clearance of both the wing and tail, the lifting surfaces will not act as a hindrance to loading/unloading personnel or equipment as they can simply move under the surfaces. Furthermore, both design choices aid in short takeoff and landing (STOL) as a high wing increases lift at takeoff and landing due to the ground effect, while a t-tail allows for more control authority during volatile phases of flight as the elevator is lifted outside of the wing blanketing zone and the engine wash.

Produced by the design process, Lifting Large determined six engines (Pratt-Whitney 4000-112) to be needed for takeoff given one engine inoperative (OEI). As will be discussed later in the report, while six engines may increase the moment arm assuming one goes out, six smaller engines were calculated to be cheaper than four bigger engines (RB Trent 877-17) by $54 million, further driving up the possible profit margin.

The final design provided by Lifting Large has not only been optimized for cost, but also for performance to greatly enhance the USAF's capabilities to spread peace throughout the world. Given the continuation of this project, Lifting Large plans to both continue aerodynamic analysis to further optimize the efficiency of the aircraft and conduct structural testing to further weight reduction and minimize the production cost of the aircraft.

Dimensioned Three-View Drawings (Aayush)

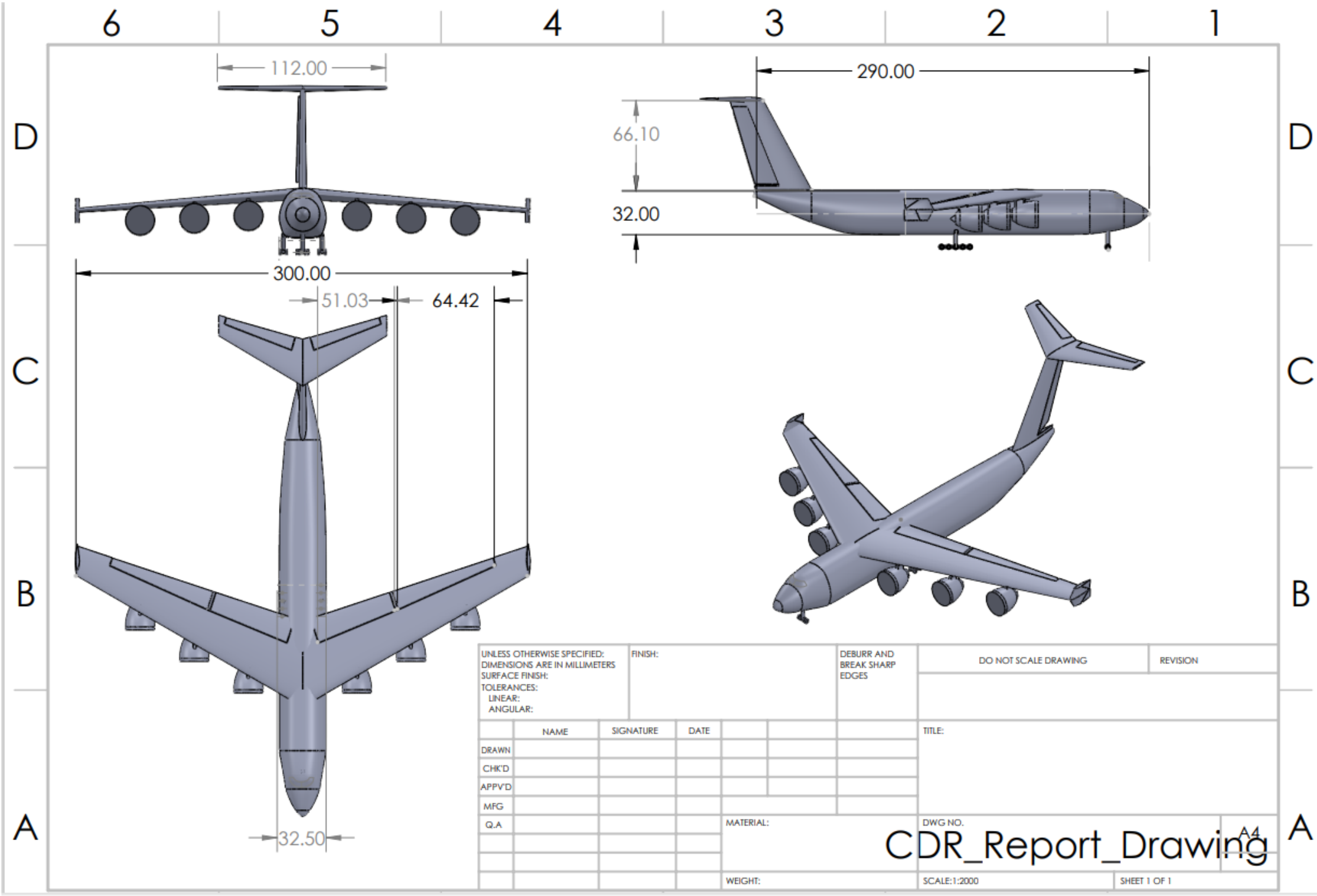


Figure 1. Dimensioned Three-View Drawings of Current Design

Major Geometric/Performance Parameters (Aayush, Rawan, Andrew)

Table 1. Major geometric and performance characteristics of the design by Lifting Large compared to other similar, existing aircraft

	Lifting Large	**C-5M**	**AN-225**	**Boeing 747 8F**
Takeoff weight [lbs]	1,771,279	840,000	1,280,000	987,000
Empty weight [lbs]	918,117	375,000	628,317	434,600
Payload [lbs]	430,000	281,000	551,155	152,400
Cruise, Landing, Takeoff CL	1.0, 2.62, 2.4	0.8, 1.0, 1.6	-	-
Cruise L/D	15	14	-	-
T/W [lbf/lbm]	0.271	0.244	0.242	0.273
W/S [lbs/ft²]	148	145	145	165
Maximum Engine Thrust [lbf]	79,960	51,250	51,590	67,400
Engine cruise SFC [1/hr]	0.5	0.47	0.35	0.55
Span [ft]	300	223	290	225
Reference Area [ft²]	12,000	5,800	9,741	5,963
Aspect Ratio	9	8.5	8.6	8.5
Average Wing t/c	10.20%	-	-	-
Cruise Mach No.	0.80	0.78	0.75	0.85
Maximum Range [nmi]	2,500	4,800	8,310	7,500
Maximum Range Fuel Burn [lbs]	421,723	-	-	-
Static Margin	24%	-	-	-
Maximum Landing Distance [ft]	7,500	4,900	11,000	11,000
Maximum Takeoff Distance [ft]	8,700	8,300	9,000	15,500

Introduction (Ian)

Appending to the requirements information noted in the *Executive Summary* of this report, this section will go into further detail on specific, quantitative requirements for the replacement aircraft, along with providing a short description of the various stakeholder needs.

Beginning with the stakeholder needs, the USAF specifically requested that minimal ground equipment should be required, with a low main deck ground clearance and ability to "drive on and off" as appropriate. Aside from acting as a driving factor for the high wing and t-tail design choices, this stakeholder need also affected the landing gear and loading ramp design, such that the maximum ramp down angle of 12° was met with a reasonable ramp length. This is further discussed in the *Landing Gear* section of this report. Finally, a minimum profit margin of 10% was desired, leading to unit production cost optimization in the form of empty weight reduction, fuel weight reduction, wing area optimization, and engine cost optimization. While these are discussed in various sections of this report, it is important to note the bulk of optimization came from overlaying an objective function (unit production cost) on the T-S plot to optimize both the thrust and wing area required for the design. This is discussed further in the *Objective Function* and *Final Sizing Plot* sections of this report.

Aside from the stakeholder needs, the aircraft needed to meet a strict set of requirements specified in the request for proposal (RFP), all of which applicable to this phase in the design are shown in the table below.

Table 2. Requirements Summary

Performance	**Payload**	**Sizing**	**Operation**
Max cruise mach of 0.82	Max payload of 430,000 lbs	Max span of 262 ft per ICAO class F footprint	Consistent with FAR 25 Balanced Field Length: 9,000 ft at ISA
Service ceiling at 43,000 ft Cruise capable at 31,000 ft	Eight total crew members (four crew and four relief)	Payload acts as main sizing requirement	Service life of 40,000 hrs
Range: ● 2,500 nmi for a full (430,000) lbs payload ● 5,000 nmi for a (295,000) lbs payload ● 8,000 nmi for a ferry flight (no payload)	Modular with three configurations: ● Three M1A2 Abrams Battle Tanks ● 48 463L pallets ● 330 fully equipped troops in the cargo bay and 100 additional fully equipped troops in separate deck	Cargo hold must be at least 13.5 ft tall, 19 ft wide, and 33 ft long (length of three M1A2 Battle Tanks in line end to end)	All weather type and incorporate deicing, terrain following and terminal avoidance systems Capable of modified mission and reserve mission profile consistent with NextGen Initiative

Interior Layout (Aayush, Rawan, Vishwa)

The interior layout consists of a cargo bay that has a main deck for cargo and passengers, and a separate deck for extra passengers. There is also detail on the cockpit's layout and pilot's sight lines.

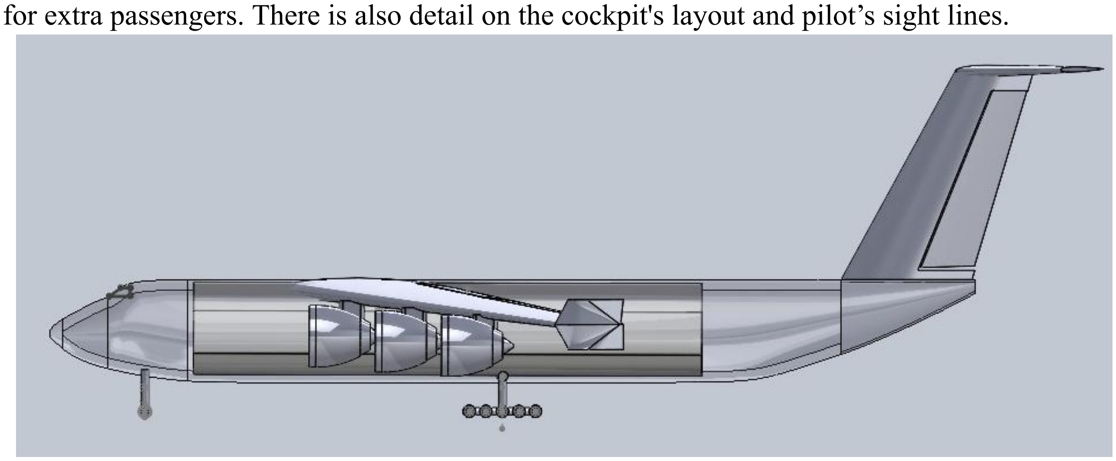


Figure 2. Cargo bay arrangement within fuselage

The Cargo Bay is designed to be 160 feet (ft) long, with the cross-section as depicted in the given figure. There are two decks, namely the main deck and the top deck. The main deck is designed to hold all the required payload and passengers, while the top deck is designed to hold extra passengers (which is the necessity for one of the RFP payload configurations). Both the decks abide by the RFP requirement floor width of 19 ft. The main deck abides by the RFP requirement of 13.5 ft floor height to store the cargo. The top deck is designed with a 9 ft floor height. The height is designed such that the Wing Spar Beam joining the wings to the fuselage (that are 3 ft wide) will leave a 6 ft height for the passengers to not get hindered by its presence. Moreover, the bottom of the Cargo Bay is designed to accommodate fuel tanks as well which will be discussed in further sections.

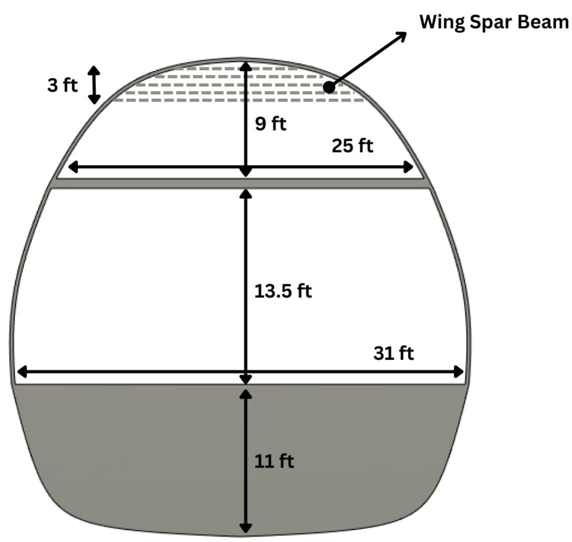


Figure 3. Cargo bay cross section

The 31 ft width of the main bay floor was made possible due to the landing gear requirements that had to be placed at least 31 ft apart from each other. Additionally, to make the belly of the fuselage aerodynamic enough to not cause any ill effects while minimizing the frontal area, the bottom of the cargo bay was designed in such a way. This enabled the cargo bay bottom to be wide enough to comfortably store the fuel tanks, and all the necessary electronics along with the landing gears and the main bay to be wide enough for extra bulky equipment to be stored by the troops and passengers, near the walls of the fuselage and for as long as the total weight of the aircraft remains under the design MTOW.

Cargo Bay Loading Configurations

Based on the requirements stated by the RFP, the cargo bay is designed to accommodate 3 different configurations of payload. Namely,

Configuration 1: Three M1A2 Abrams Battle Tanks, weighing 71.2 tons each. The tanks are spaced 18 ft apart from each other, with their centroid being at the centroid of the cargo bay.

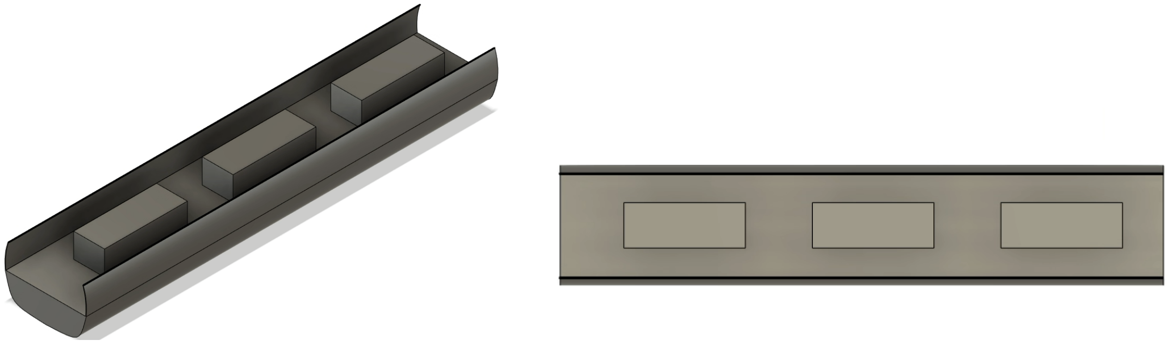


Figure 4. Loading Configuration 1 - Tanks

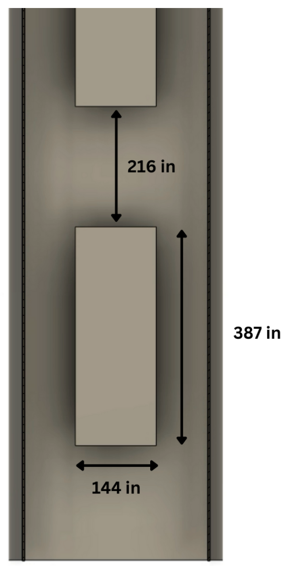


Figure 5. Loading Configuration 1 - Dimensions

Configuration 2: The second loading configuration required us to load the main bay with 48 463L pallets (a standard large air mobility pallet 88 inches wide, 108 inches long). They are arranged in a 3 column fashion, with each column having 16 pallets. The arrangement is made such that there is over 1 ft gap between each of the pallets to allow the crew to move around freely in the main bay while having a full payload. The update from PDR in the cargo bay design allows us to now fit all the pallets in the mainbay for easy loading and unloading.

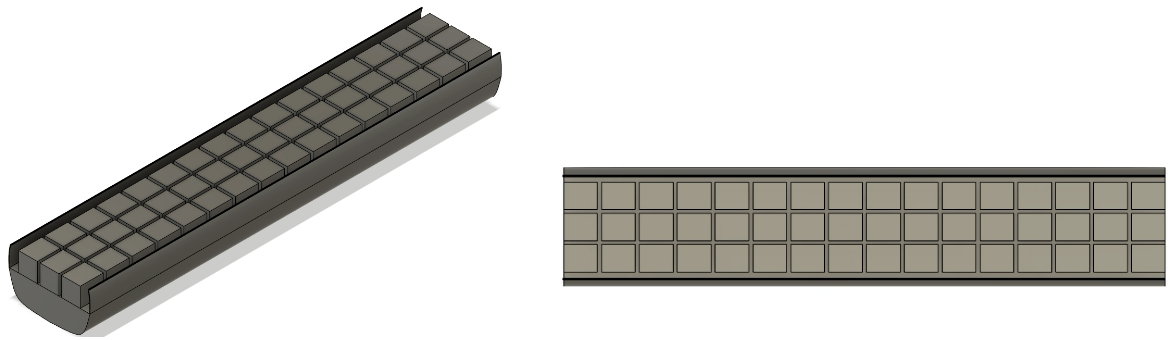


Figure 6. Loading Configuration 2 - Pallets

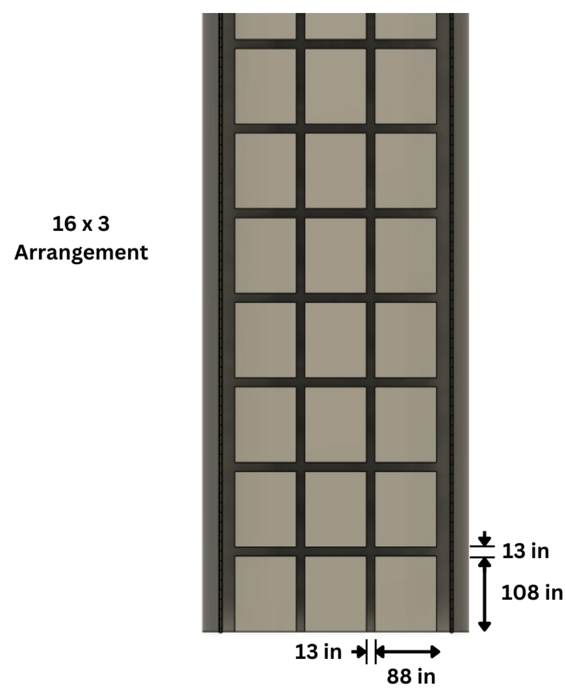


Figure 7. Loading Configuration 2 - Dimensions

Configuration 3: The third loading configuration required us to have 330 fully equipped troops in the cargo bay and 100 additional fully equipped troops or passengers in a separate deck or compartment from the main cargo bay(s) i.e. the top bay in this case. The seats are designed to have a seating space of 16x16 sq. inch. The main bay seating arrangement is designed in a five column fashion, consisting of 66 seats in each column while maintaining a 1.5 ft gap between the columns. The top bay seating arrangement is designed in a two column fashion, with 50 seats in each column while maintaining a 2 ft gap between the columns. Moreover, the extra space near the walls of the fuselage can also be used for the troops to store bulky equipment if necessary. The update in the cargo bay design (from PDR) allows the aircraft to now accommodate the troops such that they have more than enough space to seat comfortably while being able to store goods as well if necessary.

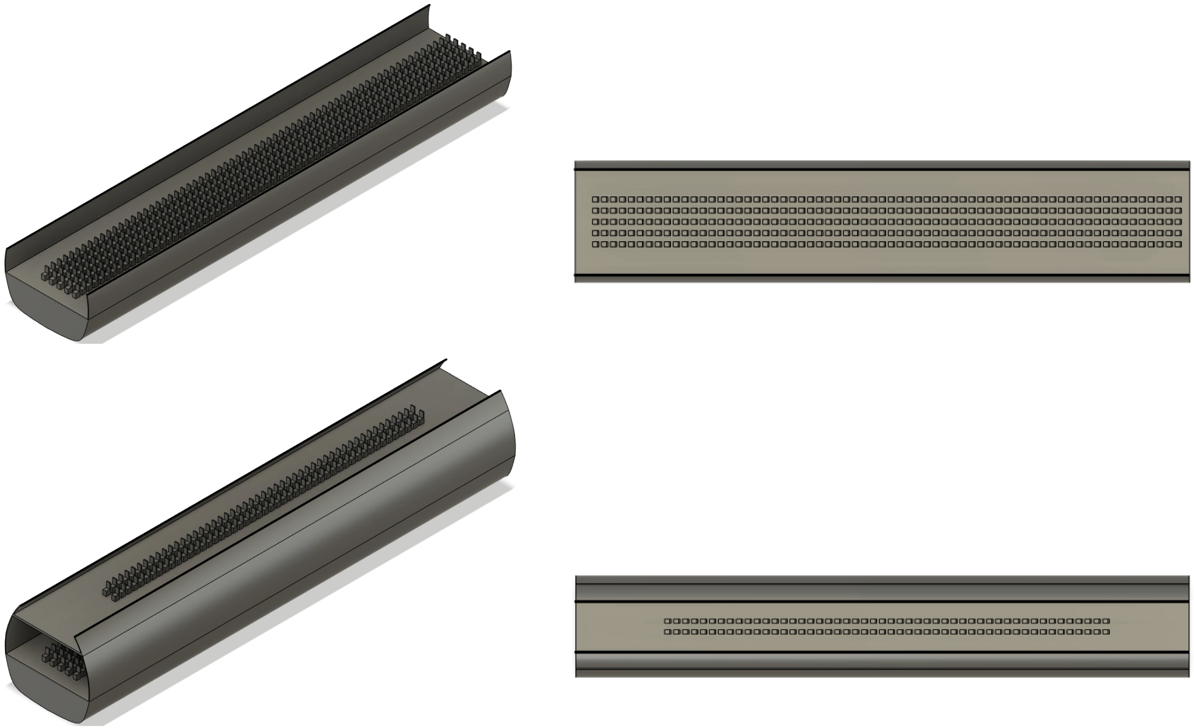


Figure 8. Loading Configuration 3 - Troops

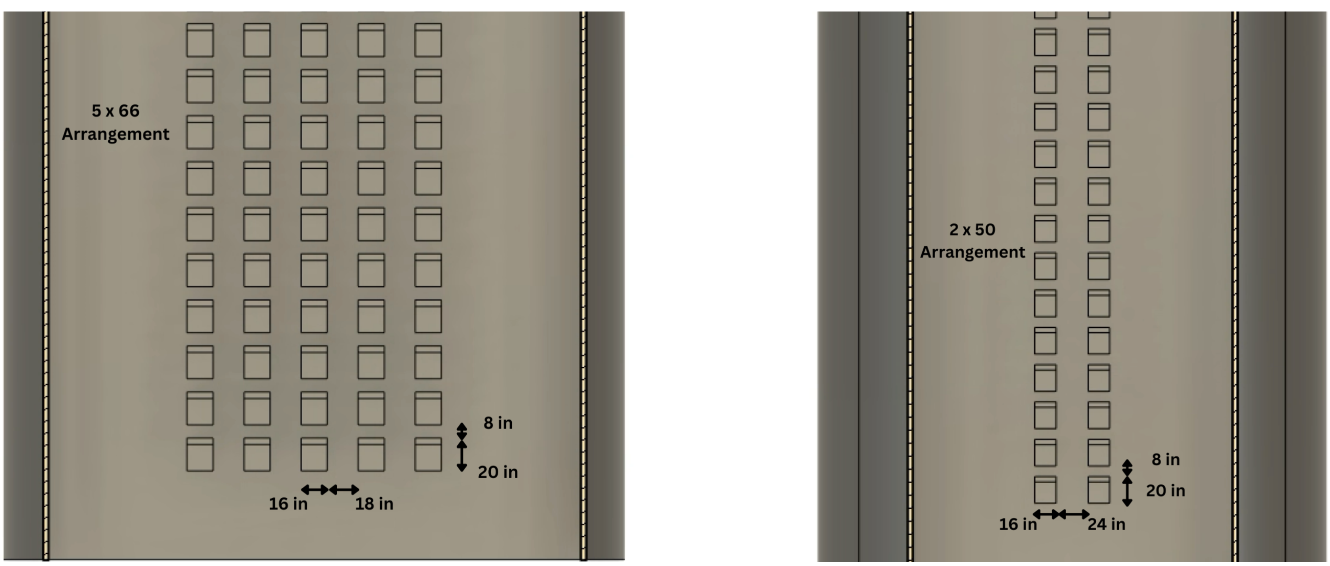


Figure 9. Loading Configuration 3 - Dimensions

Cargo Loading Ramp and Exits

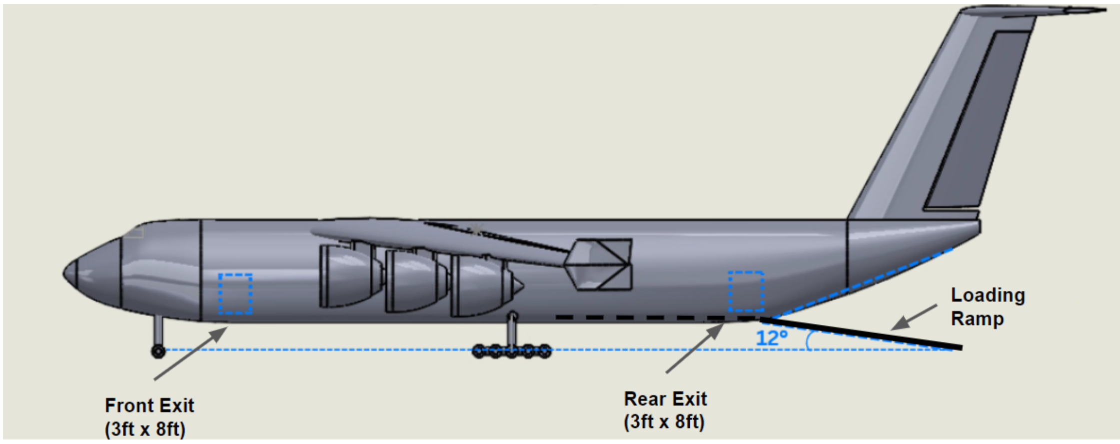


Figure 10. Emergency exits and loading ramp with incline angle

Cargo bay loading ramp and exit were designed based on the requirements from the RFP and Raymer; there are two exit gates (each being 3 ft wide and 8 ft tall) at the front and rear of the fuselage on each side resulting in four total exits. Furthermore, the ramp was designed such that it maintains a ramp angle of 12° from the ground, which abides by the FAR part 25 requirements.

Cockpit Design

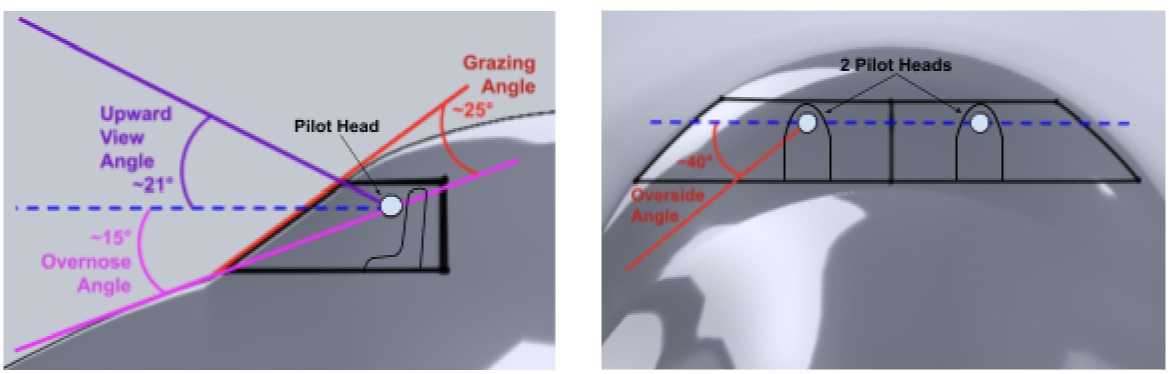


Figure 11. Cockpit Design with Labeled View Angles

There have been updates to the cockpit since the PDR Report including the windscreen size shrinking and pilot sight angles changing. Based on Raymer, the pilot should be seated around 1m from the windshield with optimal pilot visibility specifications, for bombers and transports, including a 35° downward view (overside angle) without head movement, extending to a 70° angle when the pilot's head is against the cockpit glass. A grazing angle of 30° is recommended and military standards often dictate 17° of overnose vision for military transport aircraft. Additionally, these planes should ensure unobstructed visibility extending at least 20° above the horizon for both forward and upward views. As depicted in *Figure 12*, the overnose angle is approximately 15°, the upward view angle is about 21°, the grazing angle is around 25°, and the overside angle is roughly 40°. All these angles align with Raymer's recommendations.

Fuel Tank Arrangement

The fuel tanks are designed to be at the bottom of the cargo bay, unlike the conventional arrangement in the wings of the aircraft. This was made possible due to the huge volume that the belly of the fuselage can accommodate in addition to the improved in-flight stability that this provides (discussed in further detail in the CG section). The entire volume for the fuel tanks is divided into five equal parts, each part being parallel to the fuselage length. This was done to avoid any CG imbalance in-flight while rolling the aircraft, so as to keep the fuel weight distributed evenly as opposed to being accumulated on one side of the entire tank making it difficult for the pilot to handle. The tanks are designed such that their centroid lies at the centroid of the cargo bay, to add to the stability of the aircraft. The dimensions of each of the tanks and their arrangement are provided in the following figures.

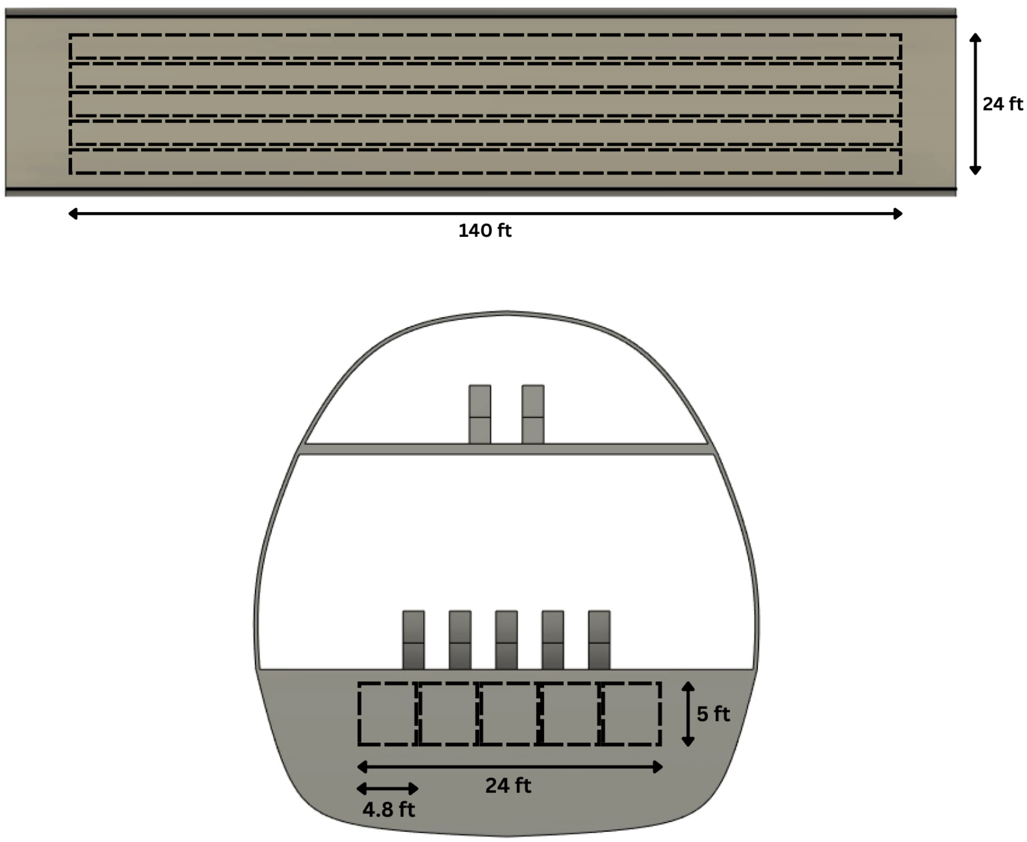


Figure 12. Fuel Tank Measurement

Aerodynamics (Ian)

The following section will discuss the aerodynamic characteristics of the design developed by Lifting Large including: the wing design, empennage design, drag polar, preliminary calculations from AVL, and finally the design refinement.

Wing Design

Beginning with the general wing sizing process, Lifting Large followed the sizing process outlined in the metabook by choosing an optimal wing area from the feasible design space in the T-S plot, discussed in detail later in the report. From the T-S plot, Lifting Large determined a wing reference area of 12,150 ft^2. Using the reference area, Lifting Large calculated a wingspan of ~300 ft assuming an aspect ratio of 7.5. The target aspect ratio was determined through comparison to similar aircraft such that the design would maintain competitive characteristics as a replacement for the C-5 and C-17 cargo transports at the EIS date of 2033.

In addition to the wingspan and reference area, the wing includes other main characteristics that aid in describing the design including the root chord, taper ratio, and sweep. Of course, the root chord and taper ratio are closely related as with a given taper ratio Lifting Large is able to determine the size of the root and tip chords given the wing area and the formula for the area of a trapezoid. Lifting Large decided on a taper ratio of 0.35, which is comparable to other cargo transport aircraft, as noted by the range of tapers given by Roskam. Then, given that the tip chord is only 0.35 times that of the root chord, the following formula may be used:

$$S_{ref} = 2[0.5(c_{root} + 0.35c_{root})(\frac{b_{wing}}{2})]$$

With a value S_{ref} of 12,150 ft^2 and b_{wing} of 300 ft, the resulting root chord of the wing is calculated to be 60 ft, meaning a tip chord length of 21 ft. Finally, sweep was chosen similarly to the wing taper, by choosing a value within the guidelines provided by Roskam. However, the chosen number was then refined using AVL to achieve a desired neutral point location for a desired static margin as will be discussed in the *Stability and Control* section of this report.

Aside from the geometric parameters of a wing, the airfoil selection plays a big role in determining the general performance characteristics of a wing. However, similar to most initial design decisions, the wing airfoil was selected given the stakeholder needs of the design acting as a replacement aircraft for the C-5 and C-17. More specifically, the RFP states a cruise mach of 0.8 with a max cruise speed of 0.82 mach. As the aircraft will be flying in the transonic regime, Lifting Large determined a supercritical airfoil to be necessary as the shape will delay the oncoming flow from transitioning to supersonic over the wing, effectively reducing drag due to the absence of shocks. With the supercritical shape in mind, Lifting Large determined the Lockheed c5a-il supercritical airfoil to best meet the stakeholder needs considering the airfoil design optimization for a very similar aircraft (C-5).

Wing Location

The wing location was determined by evaluating the general characteristics associated with each of the three possible wing locations: high, middle, and low. With the current design acting as a replacement for the currently in-service C-5, the aircraft must be able to carry very large payloads while maintaining stability in all directions (lateral, longitudinal, directional). While the wing location does not directly affect the longitudinal and directional stability of the aircraft, it plays a significant role in determining the lateral stability. Due to the class type of the proposed aircraft, Lifting Large determined that the lateral stability of the aircraft is paramount to its overall performance and mission completion, leading to a high wing design.

However, lateral stability wasn't the sole factor in determining the wing location. In addition to the benefit of increased lateral stability provided by the high wing location, the ground clearance of the wing aids in meeting the short takeoff and landing (STOL) requirement. Due to the increased ground clearance compared to other possible wing locations, the airflow around a high wing during takeoff and landing can affect the upwash and downwash of the wing, leading to a change in pressure distribution along the span of the wing. Nominally known as the "ground effect", this phenomenon aids in STOL as the effect increases lift and reduces induced drag. Additionally, the high AR of the wing allows for this effect to be more pronounced during critical phases of flight. However, given this effect changes the lifting characteristics of the wing, the pilot(s) must be aware the handling characteristics will change during takeoff and landing.

As stated by the RFP, the replacement aircraft must maintain efficient loading/unloading in the form of minimal required ground equipment and, in general, to take place over a minimal amount of time. With this in mind, Lifting Large determined the high wing location would be beneficial in allowing the landed operations around the aircraft to take place in a shorter time frame, as ground equipment may proceed under the wing in place of moving around it.

Finally, given the high wing location and the size of the wing, Lifting Large determined an anhedral would be needed to assist in the destabilization of the aircraft to aid in reaching a targeted static margin. Described in the *Aerodynamics* slide deck, a given wing location and sweep combination allows for a nominal range of anhedral values for the design. Given the table from the slides below, Lifting Large determined an anhedral of 3^{o} (dihedral of -3^{o}).

Table 3. Nominal dihedral ranges for a combination of sweep and wing position given in the *Aerodynamics* slide deck slide 17.

	Wing Position		
	Low	Mid	High
Unswept	5-7 deg	2-4 deg	0-2 deg
Subsonic swept	3-7 deg	-2 to 2 deg	-5 to -2 deg
Supersonic swept	0-5 deg	-5 to 0 deg	-5 to 0 deg

Empennage Design

The other main lifting surface of the proposed design by Lifting Large is the empennage. The design process fostered a few considerations including the tail configuration, sizing process, and airfoil selection.

The tail configuration was determined similarly to many other main design trades for the aircraft; determining what configuration would best meet the stakeholder needs and the RFP. As previously discussed in the *Wing Location* section, the empennage configuration should maintain ample ground clearance such that the loading/unloading process is efficient. At the same time, given a six engine configuration, Lifting Large determined it would be best to move the horizontal stabilizer as high as possible to both meet the stakeholder need and allow the elevator to function in clean, unwashed flow. These considerations led to a T-Tail design, a common configuration for cargo transport aircraft used by both the C-5 and C-17. On top of the increased ground clearance and elevator efficiency provided by a t-tail design compared to a conventional design, a t-tail allows for increased longitudinal stability as the elevator control authority is larger due to the increased efficiency when compared to a conventional design. While Lifting Large has not done a quantitative trade analysis between a conventional and t-tail configuration, the design choice is validated by the presence of a t-tail on most cargo transport aircraft.

Following the configuration selection, Lifting Large sized the empennage according to the tail volume coefficient method discussed in the Metabook. Given the volume coefficients for military cargo aircraft from Raymer and reducing them by 5% respectively for a t-tail configuration, the volume coefficient method was calculated using an aspect ratio of five and two for the horizontal stabilizer and vertical stabilizer respectively, as within the recommended ranges provided by Roskam. Additionally, a taper ratio of 0.44 and 0.60 were implemented, both within the recommended ranges for military cargo aircraft per Roskam. The resulting tail dimensions are shown below in *Table 3*.

Table 4. Initial tail sizing using the tail volume coefficient method including the final v-stab characteristics given by OEI analysis with size engines

	Horizontal Stabilizer (Final)	**Vertical Stabilizer (Initial)**	**Vertical Stabilizer (Final)**
Area (ft^2)	2503.0	1458.0	2186.0
Span (ft)	112.0	54.0	66.1
Root Chord (ft)	31.0	33.8	41.3

However, the sizing shown above, while final for the h-stab, is not the final for the vertical stabilizer. As will be discussed in the *Stability and Control* section, the vertical stabilizer span and root chord were increased due to one-engine inoperative (OEI) analysis given the six engine configuration (at the time of this sizing method, only four engines were considered in OEI analysis). Finally, in the same manner as the AR and taper ratio were chosen for the two stabilizers that make up the tail, the sweep was determined using recommended ranges for military cargo aircraft by Roskam resulting in a leading edge sweep of 30.4^o and 28.5^o for the horizontal and vertical stabilizers respectively.

Drag Polar

Following the completion of the main lifting surface design, Lifting Large completed a drag polar plot using the theoretical lift and drag equations described in the Metabook. Additionally, the parasitic drag was calculated and plotted according to the breakdown in the next section.

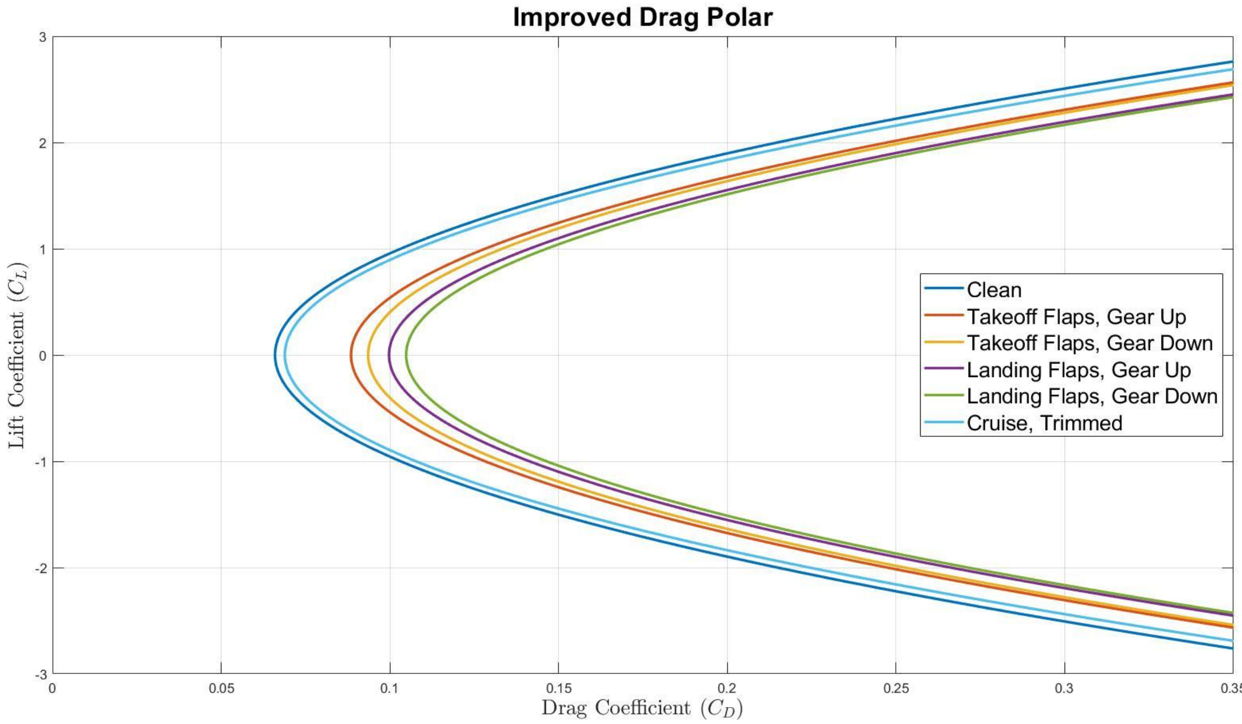


Figure 13. Drag Polar Plot with Final Design

Utilizing the drag polar plot, Lifting Large determined an absolute max L/D of 15.1, which occurs in clean cruise. During trimmed cruise, the max L/D becomes 12.6 with a calculated elevator deflection of -4.5^o at 0^o angle of attack. Furthermore, as noted in the plot above, the L/D starts to become very low as expected CL_{max} values are reached for takeoff and landing (2.4 and 2.62 respectively). While expected, the greatly increased drag during takeoff and landing may be a hindrance to the effectiveness of the pilot's control during takeoff and landing, possibly leading to extraneous circumstances in which takeoff and landing are not completed as intended. Upon continuation of the design, Lifting Large plans to take a closer look at the drag polar code, or continue to optimize takeoff and landing CL_{max} such that the distances are still within the balanced field length of 9,000 ft specified by the RFP, but don't require such large lift coefficients that lead to large drag coefficients.

Drag Breakdown

The skin-friction drag consists of the drag from the wing, horizontal stabilizer, vertical stabilizer, fuselage, and nacelles. To calculate the component drag, the equation

$$C_{D,skin\ friction} = \frac{C_f * FF * Q * S_{wet}}{S_{ref}}$$

was used. As shown in *Table 4*, The skin-friction coefficient C_f was calculated in 100% turbulent flow as predicted by Raymer for military aircraft. The form factor FF for each component were calculated as described by the geometry of the components, as well as the hinges tail configuration. The interference factors were chosen from the historical statistics for the aircraft configuration (Nacelle mounted less than one diameter away from the wing, high wing configuration, fuselage, and T-tail). Lastly, the wetted area of the aircraft S_{wet} and the wing reference area S_{ref} were obtained from the CAD geometry.

Table 5. Values calculated and assumed in the component build-up method for skin-friction coefficient, form factors, and interference factors

	Skin-Friction Coefficient	Form Factors	Interference Factors
Wing	0.000941	1.4884	1.00
Horizontal Stabilizer	0.000993	1.5448	1.05
Vertical Stabilizer	0.000955	1.5599	1.05
Fuselage	0.000768	1.2278	1.00
Nacelles	0.000986	1.2738	1.30

As shown in *Table 5* below, in addition to the skin-friction drags, the drag from fuselage upsweep was also considered in the drag component as miscellaneous form drag. This form drag takes the geometry of the fuselage, as well as the upsweep angle, to calculate the drag coefficient from flow separation.

$$C_{D,fuselage\ upsweep} = \frac{3.83u^{2.5}A_{max}}{S_{ref}}$$

For landing gears, it will only be deployed for the configuration when taking off with the landing gears, and when landing with the landing gears. For such configurations, by using Raymer's guess and the frontal area of the landing gear components, the deployed landing gears provide an additional drag coefficient of 0.0051 to the parasite drag.

$$C_{D,landing\ gear} = \frac{1.25A_{max}}{S_{ref}}$$

For flaps, the additional drags will be applied to the takeoff and landing configurations, where the flaps are deflected to generate additional lift. The flaps are designed to be slotted, with 35% of the wing chord length, and a flapped area of 5,217 ft^2. The drag from the flaps are predicted by the equation given the designed wing reference area of 12,000 ft^2:

$$\Delta C_{D0,flap} = F_{flap}(C_f/C)(S_{flapped}/S_{ref})(\delta_{flap} - 10)$$

For takeoff, the flaps are deflected by 30°, given this deflection, the takeoff flight configuration will have an additional parasite drag coefficient of 0.0225, as predicted by the equation above. For landing, the flaps are deflected by 40°, given this deflection, the landing flight configuration will have an additional parasite drag coefficient of 0.0338, as predicted by the equation above.

Furthermore, for cruise configuration, the aircraft will need to trim the flight condition, this trimmed stability will provide additional drag to the aircraft due to tail. For the current design, the horizontal tail has an aspect ratio of 5 and a reference area of 2,503 ft2. The trimmed drag can be predicted by the equation:

$$C_{D,trim} = \frac{C_{Lt}^{2}}{\pi e_t AR_t}\frac{S_t}{S_{ref}}$$

Given 3° angle of attack for trimming, from AVL results and analysis, the tail lift coefficient C_{LT} was calculated to be -0.4434, and tail efficiency e_t was calculated to be 0.90 at trimmed condition. By such, the current trimmed drag for the cruise flight configuration will have an additional parasite drag coefficient of 0.0029.

Lastly, the leakage and protuberance drags were considered by assuming an additional 3.5% of the total skin-friction drag. Furthermore, clean, takeoff, landing, and cruise configurations have different oswald efficiencies that contribute to different lift induced drag. From the analysis from AVL, it was found that the oswald efficiencies are 0.80, 0.95, 0.78, and 0.90 for takeoff, clean, landing, and cruise configurations respectively.

Table 6. Parasite drag coefficient component breakdown using finalized design with six PW 4000-112 engines, the clean configuration indicates the cruise condition without trim drag

Configuration	**Clean**	**Cruise, Trimmed**	**Takeoff Flaps, Landing Gear Retracted**	**Takeoff Flaps, Landing Gear Deployed**	**Landing Flaps, Landing Gear Retracted**	**Landing Flaps, Landing Gear Deployed**
Wing	0.0098	0.0098	0.0098	0.0098	0.0098	0.0098
Horizontal Stabilizer	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113
Vertical Stabilizer	0.0110	0.0110	0.0110	0.0110	0.0110	0.0110
Fuselage	0.0066	0.0066	0.0066	0.0066	0.0066	0.0066
Nacelles	0.0114	0.0114	0.0114	0.0114	0.0114	0.0114
Fuselage Upsweep	0.0137	0.0137	0.0137	0.0137	0.0137	0.0137
Landing Gears	-	-	-	0.0051	-	0.0051
Leakage & Protuberance Drag	0.0022	0.0022	0.0022	0.0022	0.0022	0.0022
Flap Drag	-	-	0.0225	0.0225	0.0338	0.0338
Trim Drag	-	0.0029	-	-	-	-
Parasite Drag	0.0660	0.0689	0.0885	0.0936	0.0997	0.1048

Aerodynamics in AVL

Given the final form factor of the main lifting surfaces of the aircraft, Lifting Large developed an AVL build file to further analyze the performance of the design. The following table details the results pulled from three different phases of flight: takeoff, cruise, and landing. It is important to note that the build file does not contain a fuselage, which, if included, may negatively impact a couple of the chosen performance parameters including the CL_{max} and Oswald Efficiency values. Furthermore, the build file does not include winglets in the form of a wing fence, which has recently been implemented to increase aerodynamic efficiency as will be discussed later in the *Aerodynamics* section.

Table 7. Performance characteristics of current aircraft design for different phases of flight. Takeoff and Landing phases were simulated at 10º angle of attack and Cruise was simulated at 0º angle of attack.

Configuration	CL_{max}	CM_{alpha}	Oswald Efficiency	Flap Deflection (deg)	Elevator Deflection (deg)
Takeoff	2.40	-0.12	0.80	35	10.0
Cruise	0.24	–0.13	0.90	0	-4.5
Landing	2.62	-0.11	0.78	40	20.0

Given the parameters in the table above, the aircraft remains functional and longitudinally stable in the three phases given by the pitching moment coefficient with respect to alpha (angle of attack). Using a CL_{max} of 2.62 for the stall speed calculation results in a stall speed of 127 knots (kn), on the low end for large cargo transport aircraft due to the wing size. Additionally, due to the absence of slats which will be further discussed in the *Stability and Control* section of this report, the flap deflections are on the higher end of nominal values to achieve the targeted coefficient of lift. Finally, the elevator deflections for takeoff and landing are used to increase the overall lift of the aircraft, while the elevator deflection for cruise is the trim condition.

Design Refinement

Following the aerodynamic analysis in AVL, Lifting Large decided to add winglets in the form of a wing fence to the design shown in the figure(s) below. Not only does the addition of winglets increase the aerodynamic efficiency of the aircraft, they also act as a selling point to beat out other designs. As given in the comparison chart at the beginning of this report, the design by Lifting Large has an AR of 9, larger than any of the other aircraft compared and larger than most other heavy lift aircraft.

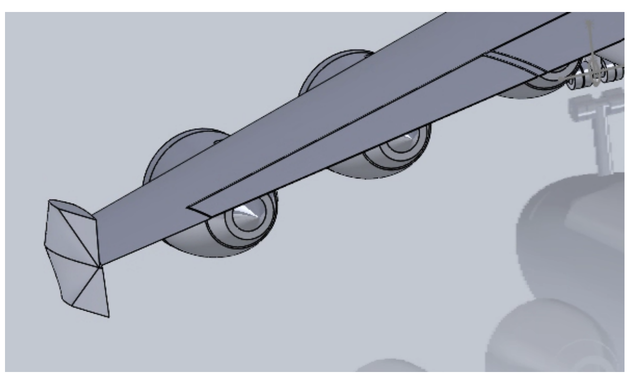


Figure 14. Detailing the wing fence implementation

However, the effect on AR provided by the wing fence is an assumption. Assuming a 20% increase in AR when adding winglets given by the *Aerodynamics* slide, slide 38, is mainly for low to moderate AR wings. Without winglets, the design has an AR of 7.5; while moderate, does not beat out the aircraft being replaced (C-5 and C-17 with AR's of 8.5 and 8 respectively). Therefore, a trade study was conducted comparing aspect ratio to fuel burn using the refined weight estimation code. The results, shown in the table below, indicate a clear decrease in fuel weight (or fuel cost) as the aspect ratio increases.

Table 8. Aspect Ratio Versus Fuel Burn Trade

Aspect Ratio	Fuel Burn (lbs)	Fuel Cost (USD)
7.5 (w/o winglets)	442,171	$342,261
8 (small increase w/ winglets)	434,058	$335,981
9 (nominal increase w/ winglets)	421,722	$326,432

Upon a quick analysis of the results of the trade study, it is clear to see why Lifting Large went with a wing fence for the final design. While the AR increase may not be as great as currently assumed, any increase in aspect ratio helps the design and decreases the direct operating cost (DOC) through fuel weight reduction, resulting in a more competitive and better performing design.

Weights and Center of Gravity (Aayush, Vishwa)

Refined weight Estimation

The detailed component-breakdown method provided in the Aircraft design Metabook is utilized to calculate the refined weight estimation of the aircraft, which is coupled with the improved fuel fraction algorithm to converge at a value of MTOW that falls in accordance with all the different component weights (some of which are dependent on the MTOW itself) and the accurate fuel fraction values.

By first separating the aircraft into parts: engine, lifting surfaces (wing, horizontal stabilizer, vertical stabilizer), fuselage, landing gear. The component weights were estimated using the approximate empty weight buildup method from Raymer, through the correlations between the wetted area of the components and the weights. Additionally, the weight of the wing was calculated using the Raymer's regression for cargo/transport aircraft.

Other weight components such as fuel, payload, and crew weights were also computed in accordance with the mission requirements, while the fuel weight was produced from the improved weight estimation code, capturing a more accurate fuel consumption by breaking down the flight stages of climb and cruise into multiple segments, each segment updates the weight and the drag it experiences. Lastly, the refined weight estimation also approximated the weight of all other components that were not considered in the code to be around 17% of the maximum takeoff weight. For such, the code loops the weight calculation until it converges, resulting in the final refined weight estimation.

The refined weight estimate is converged at :

MTOW [lbs]	MLW [lbs]	Empty Weight [lbs]	Fuel Weight [lbs]	Payload Weight [lbs]	Crew Weight [lbs]
1,771,278	1,351,712	918,116	421,722	430,000	1,440

CG estimates

The CG location of the aircraft is estimated by accounting for the weight and the moments provided by all the components while concentrating their loads at their respective CGs (the CG location for each weight component was then being calculated given the characteristic geometries of the component).

The CG of the aircraft is hence estimated to be at 118 ft (from the aircraft nose) for the maximum payload configuration. The CG locations for all payload configurations is given in the table below:

Table 9. CG Shift for All Payload Configurations

Payload Configuration	CG location (ft, from the nose)
1. 3 M1A2 battle Tanks	59.10
2. 48 463L Pallets	59.11
3. 330+100 Troops/Passengers	59.03
4. Empty Payload	58.92

CG Travel

The CG travel or the CG excursion plot is designed by accounting for the change in the load distribution in-flight for different configurations. This mainly consists of two changes that occur throughout the mission, namely the landing gear retraction and deployment and the reduction of load due to the fuel weight due to fuel usage. Four different CG distributions were calculated to find CG at pre-taxi configuration, takeoff configuration, pre-landing configuration and the landing configuration.

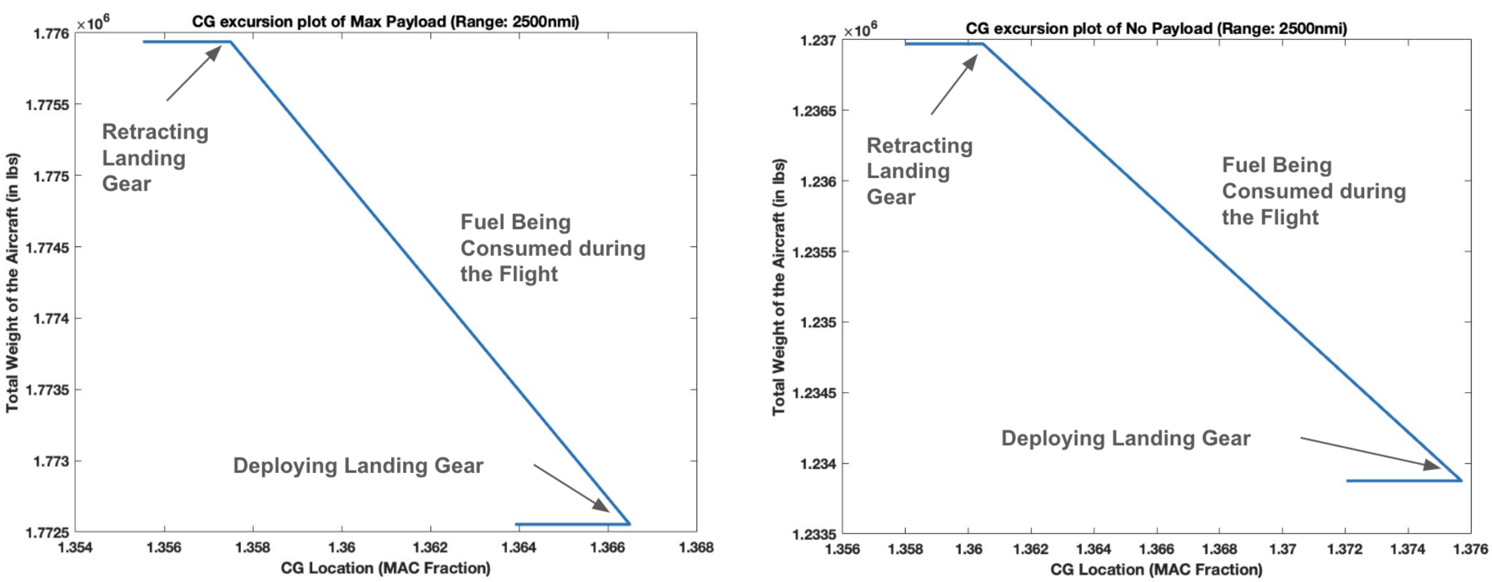


Figure 15. CG Excursion for Max (left) and No Payload (right) Configurations at Constant Range of 2,500 nmi

It is worth noting that the CG travel is extremely low for the aircraft throughout both the missions (0.01 and 0.02 MAC fractions respectively), which is made possible due to the careful designing of cargo bay and fuel tank orientations to enhance the in-flight stability of the aircraft. These plots were chosen to depict the change created due to the difference in payload weights while having approximately the same fuel due to the same range. Given below is the detailed component weight breakdown along with their CG locations.

Table 10. Detailed Weight and CG Breakdown (Component-wise)

Component	Weight (lbs)	CG Location (ft from wing root LE)
Wing	154,901	17.44
Engines (6)	97,560	41.48
Horizontal Stabilizer	9,774	225.55
Vertical Stabilizer	8,536	225.35
Fuselage	269,071	71.38
Landing Gear	76,365	49.51
Fuel	421,722	57.92
Payload	430,000	57.92
All-else empty (all components that were not accounted for)	301,909	71.38
Aircraft	1,771,278	59.10

Stability and Control (Ian)

The following section details both physical and performance characteristics associated with the stability and control of the current aircraft design.

Static Margin

Beginning with the most important stability characteristic of an aircraft, the static margin details the general stability of an aircraft based on the center of gravity location, neutral point location, and the mean aerodynamic chord of the wing. Based on Raymer, the static margin for a cargo transport should be greater than 5%. With the current design, Lifting Large is able to achieve a static margin of about 24% for all cargo configurations as supported by the small cg shift noted in the previous section for the various payload configurations. The figure and equation below detail the values used to calculate the static margin including a mean aerodynamics chord (MAC) of 43.6 ft for the wing.

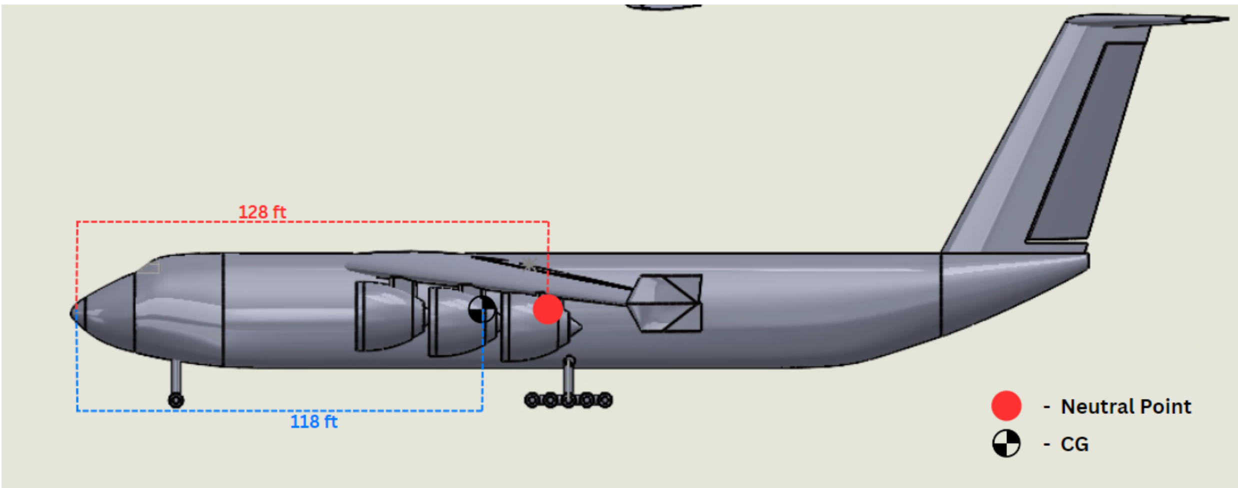


Figure 16. CG And Neutral Point Locations

$$\text{static margin} = \frac{x_{np} - x_{cg}}{\bar{c}}$$

However, Lifting Large made a few assumptions when calculating the static margin for each configuration. First, the pallet payload configuration was assumed to carry 48 463L pallets each with weight 355 lbs including side and top netting while each carrying an average load of 5,000 lbs (max load for a 463L pallet is 10,000 lbs). Second, the troop configuration cg was estimated through the assumption that an average troop with equipment weighs 248 lbs (180 lbs per person, 68 lbs of gear per person). Moving forward, Lifting Large will be cognisant of these assumptions when attempting to reach an optimized static margin, understanding that the cg may shift forward or back depending on the loading of each pallet and the specific weight of each troop and their gear.

From PDR to CDR, as mentioned in the previous report, there were a couple things Lifting Large had to play around with to reach the current static margin. First, the MAC_{wing} has increased from PDR, mainly due to the size increase of the wing from a planform area of 10,500 ft^2 to a current area of 12,150 ft^2. The change in reference area led to not only a change in the aspect ratio (8→7.5 w/o winglets), but also a change in the thickness of the wing, from a MAC of 38.9 ft to now 43.6 ft. Furthermore, as the span has also changed to 300 ft (was at 290 ft), the spanwise location of the MAC_{wing} is now at 63 ft, a little under half span (150 ft).

The other change made included shifting the wing forward and manipulating the sweep of the wing such that the neutral point was closer to the refined weight cg. The analysis done by Lifting Large in AVL resulted in a wing that is 40 ft forward from its original location on the fuselage and now has a leading edge sweep of only 30° as opposed to 40° presented at PDR.

Control Surfaces

Lifting Large currently uses a control surface layout of slotted flaps, conventional ailerons, and conventional tail control surfaces of elevators and a rudder; all of which follow the same taper ratio as their respective lifting surface (ex. Aileron and flap taper ratio of 0.35 as the wing taper ratio is 0.35). The non-dimensionalized sizing of the control surfaces is detailed in the table below. Based on control surface sizing guidelines for commercial aircraft, the flaps initially targeted a total size of 20% wing area, or 10% each. However, with the increase in wing area from PDR, this percentage has decreased to only 15%, as seen in the table below. Similarly to the flap sizing, the aileron size was determined through a rule of thumb at 12% total wing area (6% per aileron). Again, as the control surface design has not changed since PDR, the total wing area (%) of the ailerons has decreased as the wing area has increased.

Table 11. Control Surface Location and Non-Dimensional Sizing

	Flaps	Slats	Ailerons	Elevator	Rudder
Inboard span location [% of b/2]	3.4	N/A	38.5	8.6	4.5
Outboard span location [% of b/2]	40.0	N/A	84.8	91.0	90.8
Inboard chord fraction [% chord]	65.0	N/A	75.0	65.0	55
Outboard chord fraction [% chord]	65.0	N/A	75.0	65.0	55
Wing Area (%)	15.2	N/A	10.3	5.9	7.1

Given the current control surface sizing is below the rules of thumb in industry, Lifting Large ensured the new wing area percentages would still provide a controllable, flyable aircraft through analysis in AVL. The results from this analysis are noted in Table __, in the *Aerodynamics* section. The most important metric of the table of course, the CL_{max} values, were compared against nominal values for similar class aircraft (military cargo transport) provided by Roskam on slide 48 of the *Preliminary Sizing* slide deck. The nominal value ranges for military cargo aircraft consist of: 1.2 - 1.8 for clean config; 1.6 - 2.2 for takeoff; 1.8 - 3.0 for landing; with only one value provided from AVL analysis landing within the expected ranges. However, it is important to note that these are nominal ranges from aircraft already in service, none of which are as large as the current design, which may have led to the result of the CL_{max} value for takeoff being outside of the suggested range at 2.4. While takeoff and landing configs provide clearly usable CL_{max} values, the cruise config CL_{max} value of 0.24 does clearly not fit within the suggested range.

Throughout the analysis done by Lifting Large in AVL, unexpected errors occurred when attempting to load the build file, with the error message being attributed to an insufficient number of spanwise vortices. This resulted in a change of both the spacing and number of spanwise vortices on the wing to 104 and -1.1 respectively. While this may have gotten the file to load, a quick analysis of the three main phases of flight led to the discovery that elevator deflection was greatly affecting the Oswald Efficiency factor. A simple difference in clean flight between no trim and trim led to a decrease in the efficiency by about 0.25, an unreasonable amount. Given this issue was never fully understood or fixed due to time constraints working with the AVL file, Lifting Large assumes these errors to propagate into other aspects of the calculations done in AVL, possibly resulting in faulty CL_{max} values.

However, the only proponent for this case is the clean configuration CL_{max} value, as the other two are reasonable. With this in mind, Lifting Large took steps to validate the clean CL_{max} value provided by AVL. The lift equation was used at cruise conditions of 31,000 ft (density of 0.000871 slugs/ft^3, cruise speed of 795 ft/s) resulting in a lift force of only 802,619 lbs. Therefore, Lifting Large moved to fly at an angle of attack of 3^o during cruise to provide a CL_{max} value of 0.45 resulting in a lift force of 3,741,562 lbs, almost double what is needed for the current weight of the design. However, given a slight angle of attack during cruise is widespread within the industry, Lifting Large felt this calculation validated the results from AVL analysis. With an angle of attack of 3^o, the new elevator trim condition is set to -14^o, an unreasonable amount. Assuming this design continues, Lifting Large plans to optimize cruise conditions further, possibly including an angle of incidence to the elevator to mitigate the trim condition.

Another aspect of the design worth noting is the absence of leading edge slats. Lifting Large has chosen to omit this control surface for a couple of reasons. First, through extensive aerodynamic analysis into the current wing design from PDR to CDR (AVL, drag polar, updated T-S plot), the control surface sizing has proven functional as the CL_{max} values were reasonable and successfully provided a STOL under the required 9,000 ft balanced field length. Second, by eliminating a control surface from the design, its complexity, failure modes, and maintenance costs all decrease as there are less moving parts. This leads to an aircraft design that is both simple, yet competitive as the performance metrics without slats meet all related requirements. Given the finalized control surfaces of the wing, the current design is shown in the figure below.

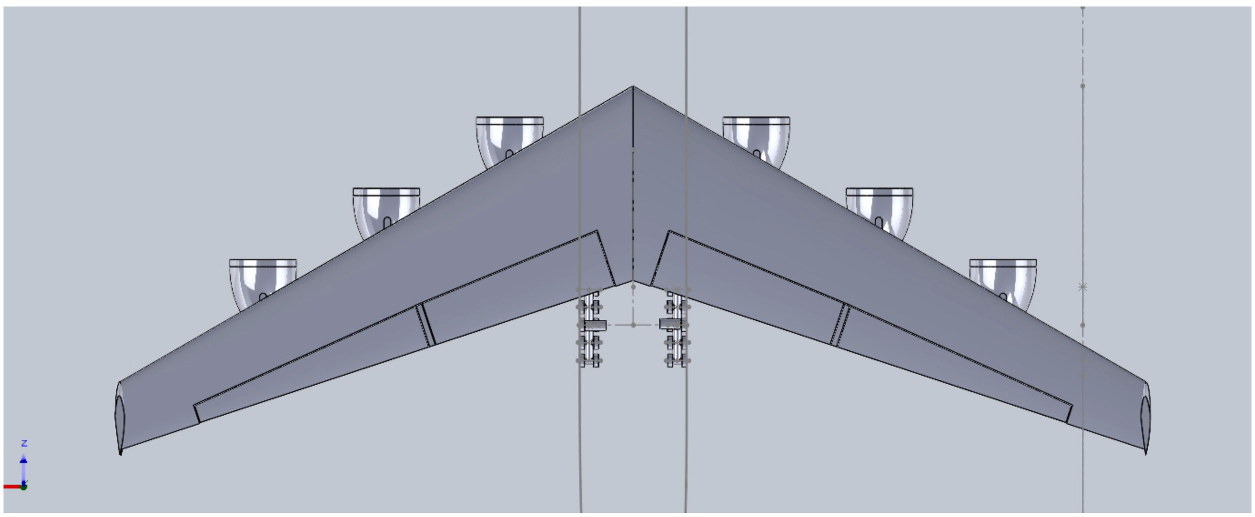


Figure 17. Final Wing Design Including Flaps (inner surface) and Ailerons (outer surface)

Contrary to the wing control surfaces, the empennage control surfaces were sized such that the desired controllability was reached. This mainly required testing in AVL to confirm both the elevator and rudder sizing. Updated from PDR, the tail control surfaces do not extend the entire span of the stabilizer, rather start just after the beginning of the stabilizer and end a bit before the stabilizer. This allows space for the two stabilizers to be structurally attached to one another, before attaching the t-tail configuration to the fuselage. Additionally, it is important to note that the elevator size has not changed from PDR and still maintains 35% chord length along the vertical stabilizer. This is due to the horizontal stabilizer size remaining constant since the initial design before PDR. However, as noted in the *Aerodynamics* section, the vertical stabilizer and rudder size underwent a resize due to the tail volume coefficient method sizing unable to meet the OEI case for six engines.

Assuming worst case OEI (engine farthest out from fuselage centerline goes out), the rudder was sized to enable controllability during cruise with an engine-out moment of 12,500,000 ft-lbs. Following the procedure outlined in the *Weights and Balance* slides, the rudder must be able to counteract this moment with a max deflection of 25^o. With a calculated minimum control speed of 257 ft/s ($1.2V_{stall}$) and a horizontal moment arm of 165 ft, the resulting moment at a max rudder deflection is only 1,700,000 ft-lbs with the updated rudder and vertical stabilizer.

Given time constraints near the end of the semester, Lifting Large was unable to find any error in this calculation relating to equations used, units used, or input values. However, the vertical stabilizer design is already much larger than nominal for passenger aircraft, and still proportionally larger than the C-5 vertical stabilizer. Therefore, it follows that the design should still be functional, regardless of the calculation failing at 257 ft/s (minimum control speed). In an attempt to validate the current vertical stabilizer and rudder sizing, Lifting Large completed the moment calculation constraining the moment provided to the OEI case. This resulted in a required minimum control speed of 699 ft/s, about 200 ft/s less than the nominal cruise speed.Given this calculation, assuming one engine goes out during cruise, the aircraft is capable of maintaining control and executing an emergency landing. Additionally, assuming an engine goes out during takeoff, the aircraft has enough time to stop within the required field length such that a takeoff is not needed. In the event an engine goes out below the minimum control speed of 699 ft/s, the aircraft pilot would need to reduce the strength of the remaining engines to be below takeoff thrust in order to maintain control authority in the rudder, a common technique for emergency scenarios. Finally, the calculation assumes a worst case scenario, in which each engine is providing max thrust. For the majority of the flight this will not be the case and the required moment from the rudder will drastically decrease after initial takeoff and climb.

Moving forward with the design, Lifting Large plans to investigate the OEI rudder sizing in further detail, possibly using AVL to validate the current sizing with induced moments. Additionally, the number of engines and engine locations could be altered to mitigate the moment caused by OEI, possibly leading to a smaller vertical stabilizer and rudder design.

Finally, given the fully updated empennage design, a figure is shown below detailing the design.

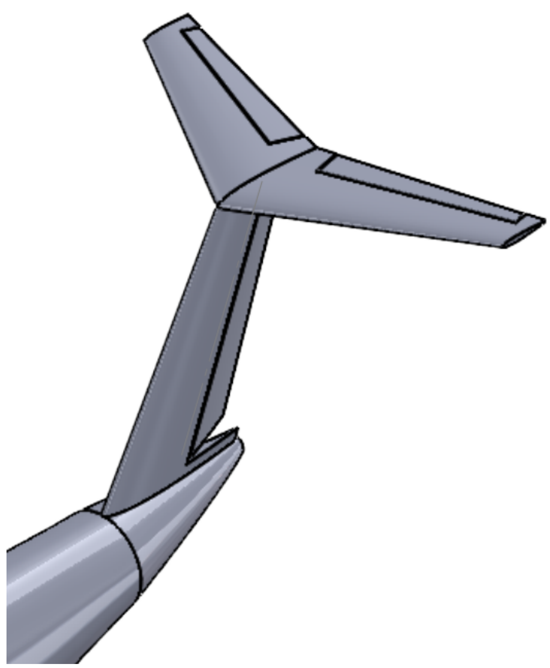


Figure 18. Finalized Empennage Design Including Control Surfaces

Propulsion System (Rawan)

The maximum required thrust for the Lifting Large aircraft is approximately 470,000 pounds of force (lbf). This thrust value has been revised since the Preliminary Design Report (PDR) due to many design changes since PDR as discussed throughout this report, including a reevaluation of the overall weight of the aircraft. The amount of engines has also increased from four to six as employing four engines would require each engine to have at least 117,500 lbf of thrust, which is currently not on the market. So, the six engine configuration was adopted with a minimum of 78,334 lbs of thrust each. The engines are to be mounted through pylon struts connecting the engine nacelles to the wings. The fuel is derived from the bottom of the fuselage as explained previously in the Fuel Tank Arrangement sub-section in the Interior Layout section.

The aim is to have engines that slightly exceed the minimum thrust requirement to ensure there is adequate power without being excessively overpowered. There was a design trade done with two engines, and a theoretical third engine which was included only for conceptualization. The third engine is theoretical because there is no off-the-shelf engine readily available with a thrust of 117,500 lbf or higher.

Table 12. Engine Comparison

Engine	PW-4000-112 [Selected]	RB-Trent-877-17	Theoretical
Manufacturer	Pratt & Whitney	Rolls Royce	-
Type Certification Year	2012	2019	-
# of Engines Implemented	6	6	4
Thrust (lbf)	79,960	78,908	117,500+
Total Thrust (lbf)	479,760	473,448	470,000+
Specific Fuel Consumption	0.5	0.5	-
Cost per Engine	$16M	$25M	Most Expensive
Total Cost (Qty)	$96M	$150M	Most Expensive
MTOW (lbs)	1,771,279	1,769,549	-
Bypass Ratio	6.2	6	-
Other Aircraft Utilized	Boeing 747, 777	Boeing 777	-

As shown in *Table 11* the two engine options that were seriously considered were the Pratt & Whitney 4000-112 and the Rolls Royce Trent 877-17. Expectations for how some parameters would look like for a theoretical engine that would satisfy a four engine configuration was also documented for the sake of consideration. Ultimately, what set the PW-4000-112 apart was its significantly cheaper price despite highly similar parameters like thrust, efficiency, and bypass ratio to the RB-Trent-877-17.

Landing Gear (Aayush, Ian)

The landing gear sizing was completed using two sources. The first, an aircraft design book by Mohammad H. Sadraey, contains the full sizing process in chapter 9. The other source used was Raymer (2018) to size the wheels.

Assuming a tricycle configuration is chosen, there are multiple modifiable parameters that make up the landing gear sizing specifications. First, and most importantly, is the fuselage ground clearance. The fuselage ground clearance directly relates to the clearance angle, tip back angle, and the overturn angle. Fortunately, the fuselage ground clearance has already been indirectly determined by the RFP. Per the AIAA design competition, the aircraft cargo loading ramp must have no more than a 12 degree inclination angle with the ground. By assuming a ramp length based on the current aircraft fuselage taper, it is possible to solve for the max clearance height of the fuselage with some trigonometry shown below.

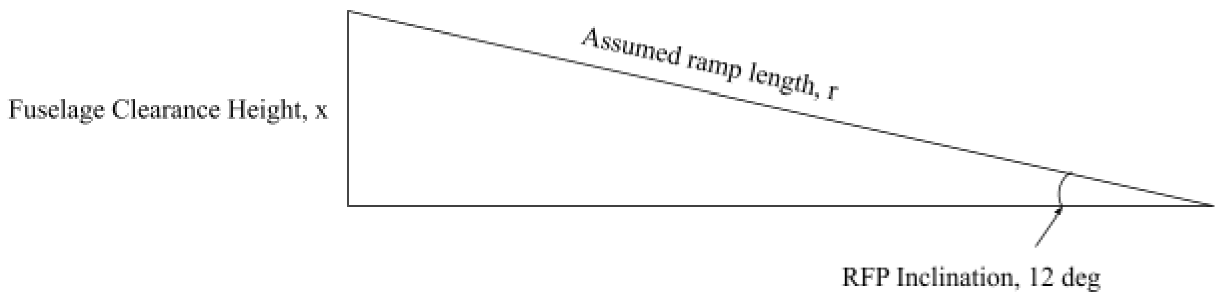


Figure 19. Max Fuselage Clearance Height Derivation

With an assumed ramp length of r = 47 ft, the max fuselage ground clearance is calculated to be 9.77 ft. With a targeted fuselage clearance and the main gear beginning 50 ft in front of the beginning of the fuselage taper, the resulting clearance angle is about 10^o, denoted by a blue dashed line in the figure below. The next parameter important in landing gear calculations is the wheelbase, or the distance between the nose and main gear in a tricycle configuration.

Depending on the wheel base, the main and nose gear shift relative to the cg location of the aircraft, determining the static load on both parts. As recommended by Raymer, the desired static loading on the gear is split 15% to 85% between the nose and main gear respectively. However, this was not the only factor in determining the wheel base. As recommended by Sadraey, the desired tip back angle of an aircraft is 5 deg greater than the takeoff rotation angle. This will not allow the aircraft to tip back onto its tail in most nominal conditions aside from incorrect cargo loading or extreme weather conditions. For a military cargo transport aircraft, it is assumed that the takeoff rotation angle is around 10 deg angle of attack. This would mean a target tip back angle of 15^o. To accomplish this, a parameter derived from the wheelbase is used; the distance from the cg to the main gear. Simply put, as this distance becomes greater (as long as the main gear is behind the cg), the tip back angle becomes greater. However, this must be traded with the wheelbase to ensure proper static loading on the nose wheel as if the main gear is too far back, a substantial portion of the static load will fall on the nose wheel rendering the design infeasible. Another important parameter of the tip back angle is the cg height relative to the ground. Currently, the cg height is estimated to be the height of the fuselage clearance added to the cg location within the fuselage which is currently assumed to be at half the fuselage height, or 17.34 ft. Again, with some trigonometry described below, the tip back angle was calculated to be 21.7^o, exceeding the target angle by greater than 6^o.

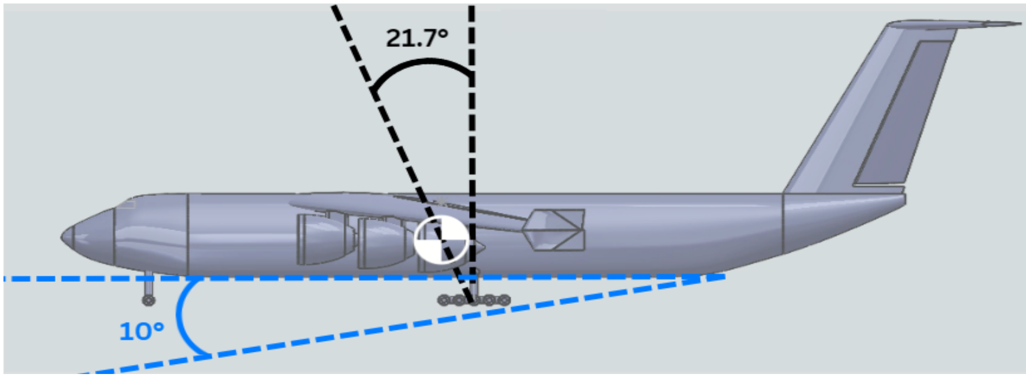


Figure 20. Finalized Clearance (blue) and Tip Back (black) Angles

Finally, the last important parameter that has not been discussed is the wheel track. The wheel track measures the distance between the two sides of the main gear in a tricycle configuration. As noted in the figure below, the wheel track is directly related to the overturn angle of the aircraft which is recommended to be greater than 25º by Sadraey for the described angle in the figure, or about 65º for the angle described in the metabook, denoted in the figure below by a red line.

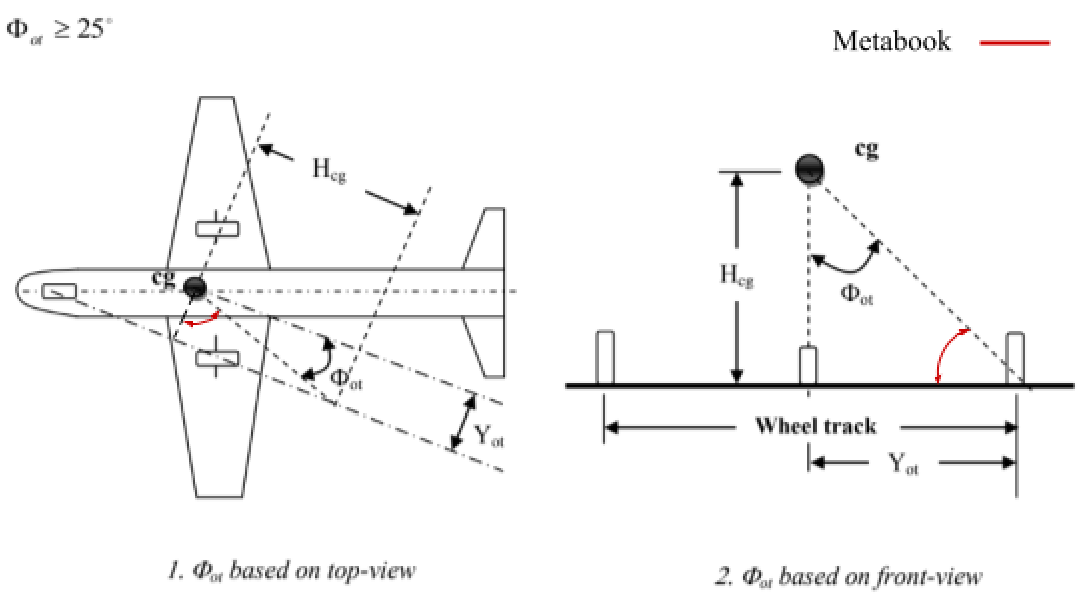


Figure 21. Overturn Angle Definition

From the above figure, unsurprisingly, trigonometry can again be used to derive a relationship between the wheel track, CG height with respect to the ground, and the overturn angle. Using the equation below, the overturn angle can be constrained to greater than what is recommended by Sadraey at 30º. This is simply to maintain greater stability during taxi as the design aircraft is a massive cargo transport vehicle made to carry very large payloads.

$$tan(\phi_{ot}) = \frac{Y_{ot}}{H_{cg}}$$

With values to describe both the overturn angle and cg height (30º and 27.1 ft respectively), one may solve for the distance between the cg and the main gear wheels. Per Sadraey, the wheel track must be at least twice Y_{ot} to avoid overturning from wind conditions, leading to a total wheel track of 31.3 ft. The resulting overturn angle as described by the Metabook is shown in the figure below.

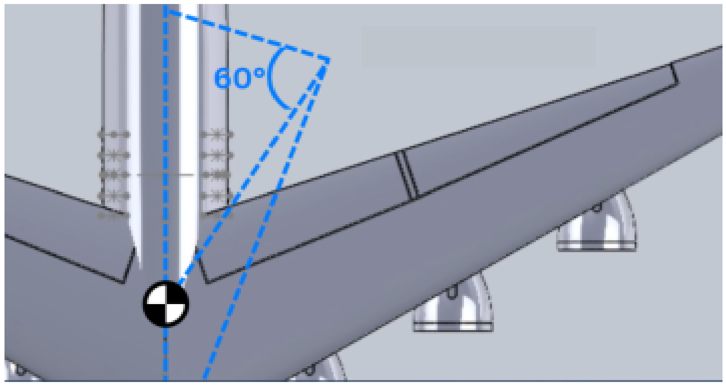


Figure 22. Finalized Overturn Angle

Wheel Sizing

Wheel sizing was derived from chapter 11 in Raymer 2018, specifically, table 11.1 shown below. With the weight of the wheel being derived from the total static load and the amount of wheels, one must simply pick the parameters out of the table for the desired aircraft class and follow the given formula.

Table 13. Wheel Sizing by Raymer

Aircraft Type	Diameter		Width	
	A	B	A	B
British units: Main wheels diameter or width (in.) = AW_W^B				
General aviation	1.51	0.349	0.7150	0.312
Business twin	2.69	0.251	1.170	0.216
Transport/bomber	1.63	0.315	0.1043	0.480
Jet fighter/trainer	1.59	0.302	0.0980	0.467
Metric units: Main wheels diameter or width (cm) = AW_W^B				
General aviation	5.1	0.349	2.3	0.312
Business twin	8.3	0.251	3.5	0.216
Transport/bomber	5.3	0.315	0.39	0.480
Jet fighter/trainer	5.1	0.302	0.36	0.467

W_W = Weight on wheel.

Through some research, Lifting Large determined 24 wheels to be a suitable number for the size of the current aircraft with the static load determined in the previous section to be 15% on the nose wheels (4) and 85% on the main gear (20). Using a gross takeoff weight estimate of 1,800,000 lbs, the load on each nose wheel was determined to be 67,500 lbs while the weight on each main gear wheel is 76,500 lbs. A and B coefficients were set using the table for transport/bomber aircraft in British units. This led to the following specifications for the nose and main gear wheels:

Table 14. Finalized Wheel Sizing

Gear	Diameter (in)	Width (in)
Nose	54.1	21.7
Main	56.3	23.0

Furthermore, Lifting Large developed a preliminary method to store the landing gear within the fuselage during flight shown in the figure below. Given a track width of 31.3 ft, the base of the fuselage increased in width to allow for the landing gear to connect to the fuselage as if it were connected to the aircraft on the wing or any other way, the static load on either the strut or the component it is attached to will be too much to handle, resulting is complete failure.Fortunately, given the broad base of the fuselage, it has a width about three times in length as the fuselage clearance, allowing for the main gear to simply fold in and upwards into the fuselage. Additionally, the nose gear will retract backwards and up into the nose as a forward retraction would be impeded by the nose taper.

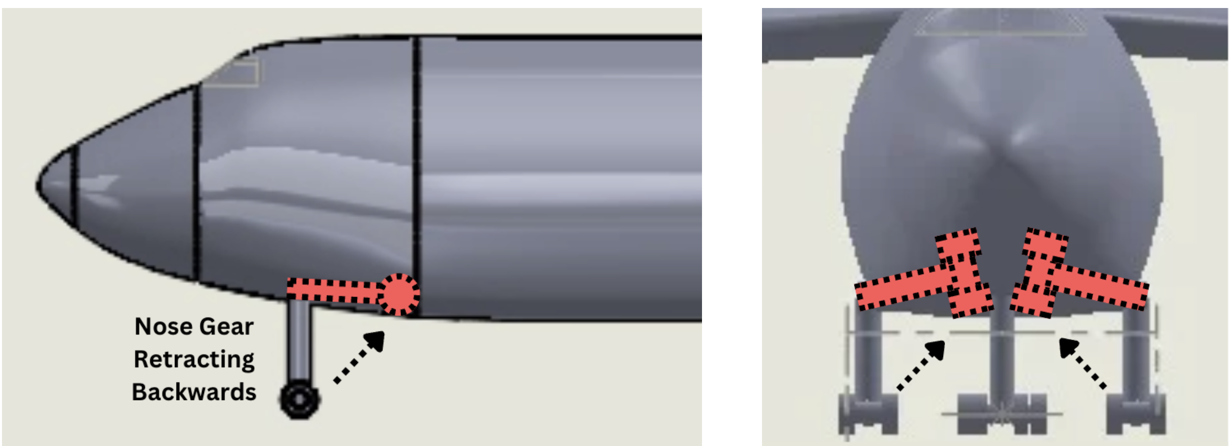


Figure 23. Landing Gear Retraction

Method Validation (Andrew)

To validate the software method, the actual data of the similar aircraft were used in the code to find the design parameters. The calculated and estimated design parameters are then being compared with the actual design parameters of the aircraft.

Weight Estimation Validation

The estimated design parameters by the code and the actual design parameters of Lockheed C-5 Galaxy are being shown in *Table 14* below. The comparison chart shows a good match between the actual data and the estimated values, with minor discrepancies. One noticeable difference is the fuel weight given by the MATLAB code and the actual fuel weight, with about 50% difference. This considerable difference is caused by the assumptions in the improved weight estimation code, such that each mission flight stage uses historical fuel fraction, and the cruise stage calculates the fuel fraction with assumptions about the drag and lift coefficients. Overall, the estimated design parameters match well with the actual design parameters. For example, the maximum takeoff weight has an estimated deviation of less than 0.1% difference from the actual data.

Table 15. Software Validation Using C-5 Galaxy; The MTOW and Empty Weight were calculated by the refined weight estimation code. The Fuel Weight and Max Lift-to-Drag Ratio were calculated by the refined fuel weight estimation code. The Cost of the Airplane was calculated by the cost estimation code.

	MTOW (lbs)	Empty Weight (lbs)	Fuel Weight (lbs)	Max Lift-to-Drag Ratio	Cost of Airplane (USD)
Actual	840,000	375,000	184,000	14	$289 million
Estimated	839,462	274,556	282,466	15	$249 million
Percentage Difference	0.064%	26.8%	48.8%	7.14%	13.8%

Cost Estimation Validation

The cost was estimated using the historical statistics regression equation provided for commercial jet transport aircraft. This equation is intended for the aircraft with maximum takeoff weight range between 60,000 lbs and 1,000,000 lbs. Thus, since the Lifting Large aircraft has a maximum takeoff weight of over 1.7 million pounds, the equation must be validated first with similar transport cargo aircraft that have MTOW close to or above 1,000,000 lbs. The cost comparison between actual and estimated data are shown in the table below. Overall, the regression equation is roughly a good approximation of the unit production cost. For example, An-225 has a maximum takeoff weight of 1.3 million pounds, with the estimated cost having only 8% difference from the actual cost.

Table 16. Cost Validation With Similar Aircraft; The actual production cost is either researched or estimated by dividing the sell price by 1.3, to account for the usual 30% profit margin of the aerospace industry. The estimated production cost is produced by using the regression equation with the maximum takeoff weight

	C-5M	An-225	BG-747-8F
MTOW	840,000 lbs	1,300,000 lbs	987,000 lbs
Actual	$289 million	$385 million	$321 million
Estimated	$249 million	$354 million	$284 million
Percentage Difference	13.8%	8.05%	11.5%

Objective Function (Andrew)

The objective function is the estimated cost of the aircraft given the desired cost escalation factor and the maximum takeoff weight of the aircraft, it does not take into account the actual price of the engine. For example, the actual cost of the engine is around \$96 million, but the estimated engine cost was around \$140 million. This difference in price was accounted for in the *Cost* section. The objective function is described as the contours in the T - S plot, an example can be seen in *Figure 24*'s background contours.

The component of the objective function is a general regression formula from the historical statistics for commercial jet transport aircraft, the valid maximum takeoff weight range for this regression is between 60,000 and 1,000,000 lbs. Therefore, since Lifting Large has a weight over 1.7 million pounds, the regression formula must be validated first before using as the estimation. The cost method validation was conducted above in the "Method Validation" section.

The estimation formula used is:

$$C_{aircraft} = 10^{(3.3191\ +\ 0.8043\ log_{10}(MTOW))} * CEF$$

The major contribution of this objective function is the maximum takeoff weight, as the cost escalation factor is fixed given the cost value conversion between different years.

By running this regression formula with the aircraft's designed maximum takeoff weight, and the cost escalation factor from year 1989 to year 2023, the Lifting Large unit production cost was estimated to be about \$454 million (before accounting for actual engine cost).

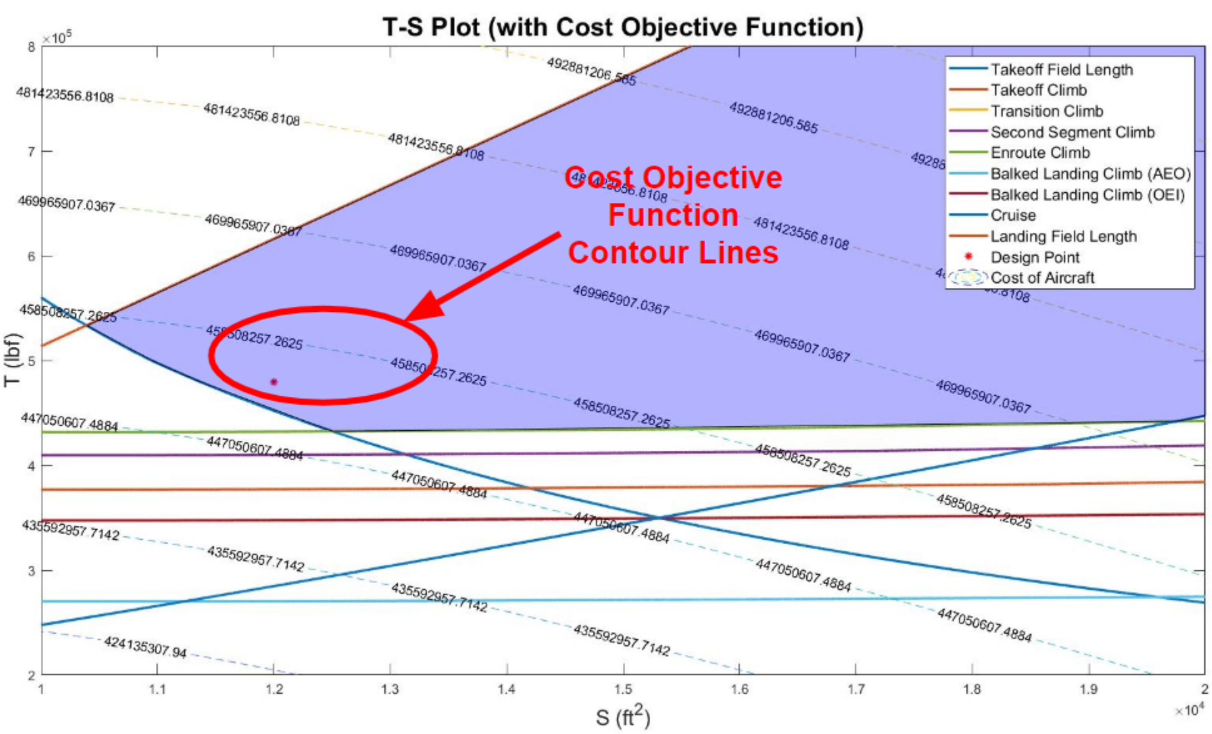


Figure 24. T-S Plot Emphasizing Objective Function Contours; This function allows the design to be chosen at a more cost effective point, therefore driving the cost of the aircraft to an optimal range

Final Sizing Plot (Andrew)

After a thorough design cycle and refinement, the final design was chosen. With a thrust-to-weight ratio of 0.27, a wing loading of 147.6 ft², a maximum takeoff thrust of 479,760 lbf, and a wing reference area of 12,000 ft². The final sizing plots are being shown below as a T/W - W/S plot in *Figure 25* and a T - S plot in *Figure 26*.

Note that for both design plots, the design point is not at the most optimal design area, that is due to the trade off between the complexity of implementing a larger wing and the cost of affording a higher thrust engine type. As stated in the RFP requirements, the aircraft must have a max span less than or equal to 262 ft per ICAO class F footprint. To accomplish this requirement with the current wing design span of 300 ft, the aircraft is implemented with the folding-wing system (used on the Boeing 777x), where the folded-wing configuration would have a span less than 262 ft due to foldable wingtip sections of 20 ft per half span.

In order for the design to reach the most optimal point, the wing loading must decrease and the wing reference area must increase. With such, the geometry of the lifting surface would require additional modifications, as well as the folding mechanism. The modification of the wing increases the total weight, and therefore the cost of the aircraft per the cost estimation code. This increase in cost, when compared to the current design with a higher-thrust engine, resulted in a more expensive aircraft production cost, while the direct operating cost remains about the same. Therefore, with the tradeoff between the production cost and the direct operating cost, a larger thrust-system was preferred over a larger wing-body, which has an overall lower cost in comparison.

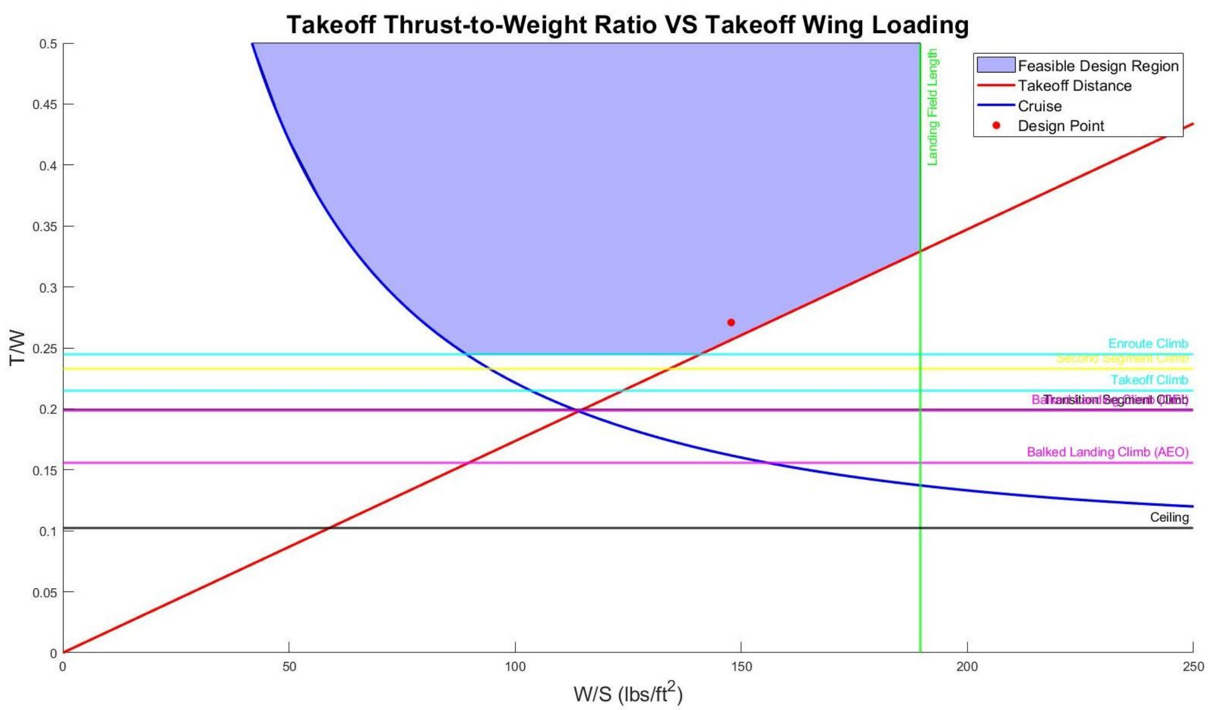


Figure 25. Finalized T/W Versus W/S Plot; The red dot indicates the final design point (T/W = 0.27, W/S = 147.6 ft²).

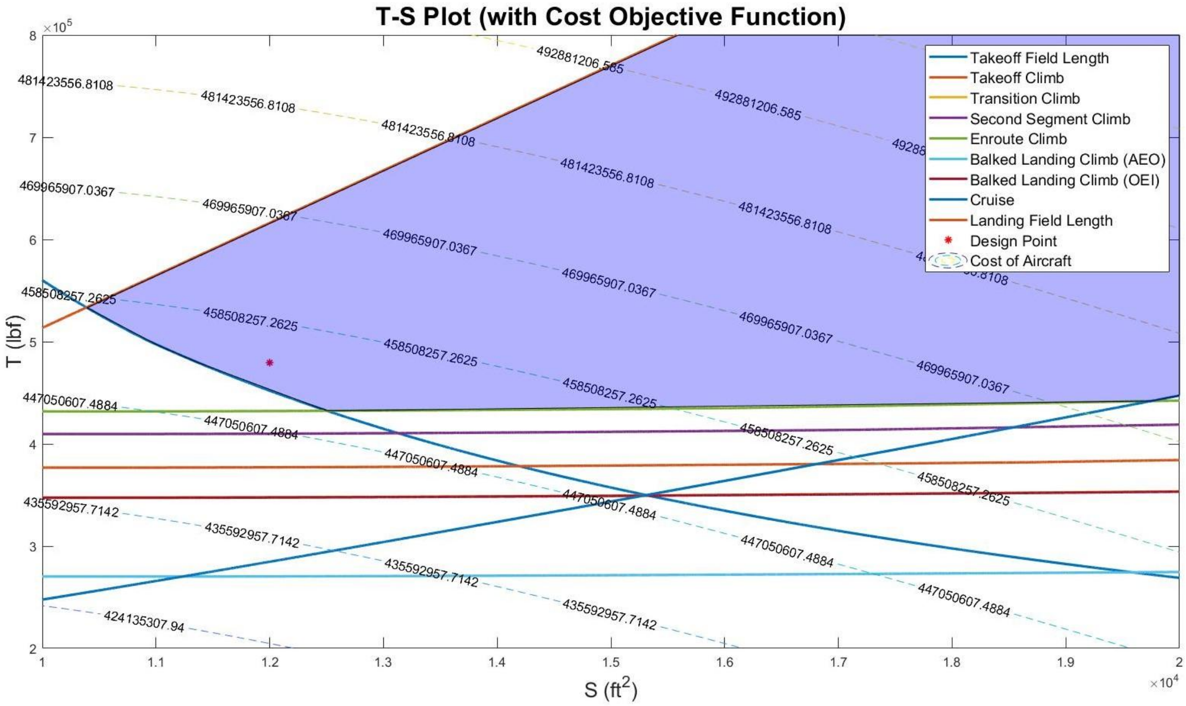


Figure 26. Finalized T-S Plot; The red dot indicates the final design point (T = 479,760 lbf, S = 12,000 ft²), optimized for cost

Structures (Ian, Vishwa)

Load Paths and Lift Distribution

The load paths on the aircraft when seen in the frontal view will be as shown in the figure below:

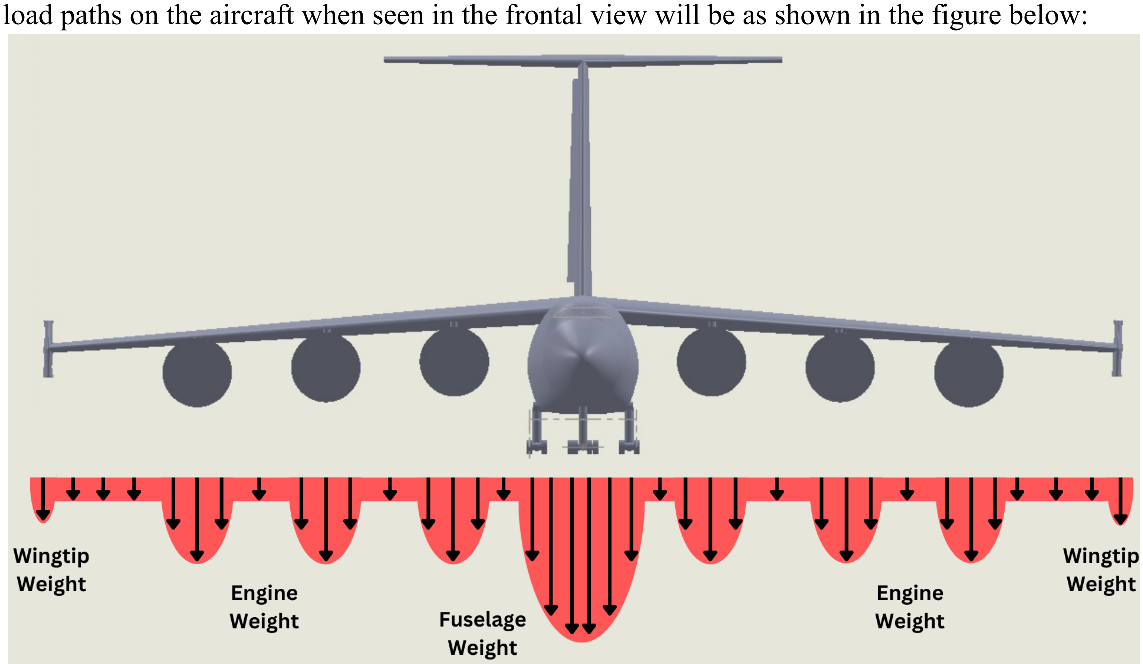


Figure 27. Nominal Spanwise Load Distribution

As shown in this image, the fuselage will be the heaviest load that the wings will have to carry. The various loads present inside the fuselage will be held with the help of longerons and stringers at the periphery of the fuselage, which will be held together by the bulkheads present at regular intervals. The entire weight of the fuselage will then be transferred to the wings with the help of the bending beam or the wing spar beam that passes through the top bay and hence through the top of the fuselage, joining the fuselage to the wings. The wings are strengthened by the span-wise spars that carry loads from several wing ribs that are situated at regular intervals inside the wings. Similarly, the engines will also be held by chord-wise wing ribs such that the load is transferred easily without damaging the wing skin which is of great aerodynamic importance.

The winglets and the engines will provide particularly higher loads on the wings due to their weight. The fuselage weight will be the highest load on the load paths. Similarly, the lift distribution is affected by the presence of extraneous structure around the wing. However, as Lifting Large obtained the lift distribution from AVL and the build file did not include a fuselage or winglets, the distribution is purely from the wing. The following figures describe the lift distribution along the spanwise axis of the wing for cruise and landing (denoted by solid green line). Takeoff has been omitted as it closely follows the landing simulation (see Appendix). Refer to the figures for the flight parameters input for each simulation.

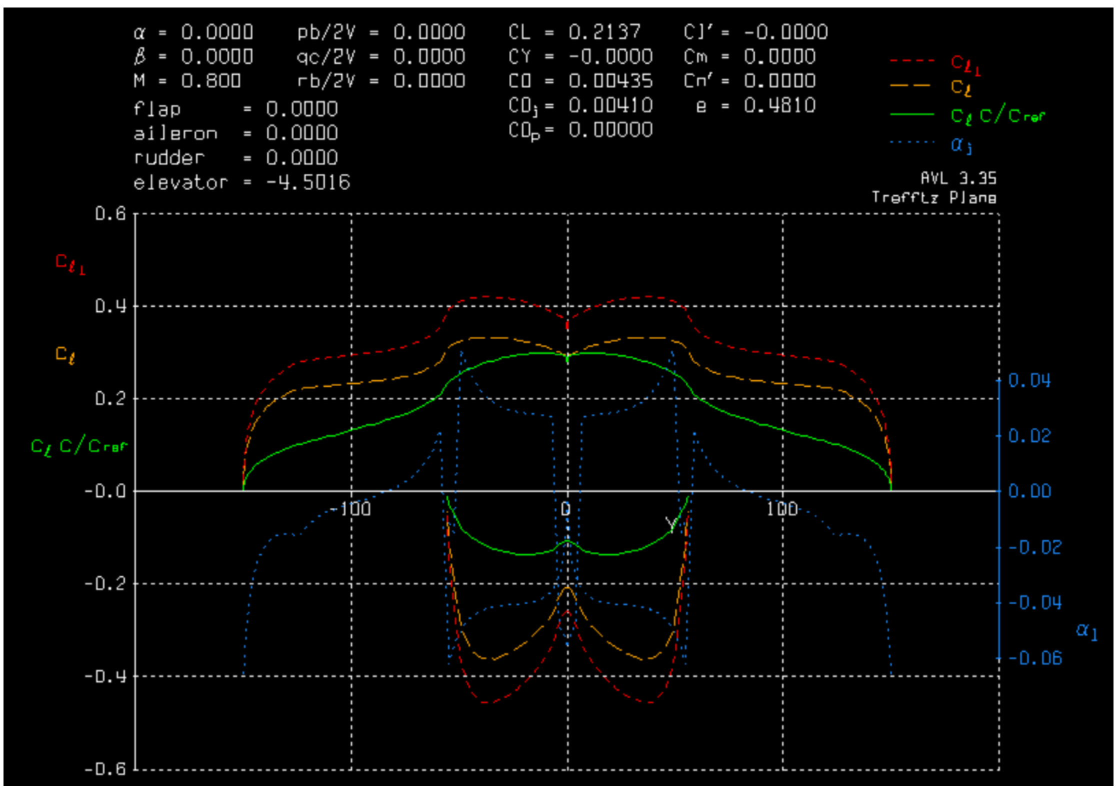


Figure 28. Lift Distribution During Cruise; The poor efficiency factor stems from error in AVL calculation discussed in the *Stability and Control* section of the report.

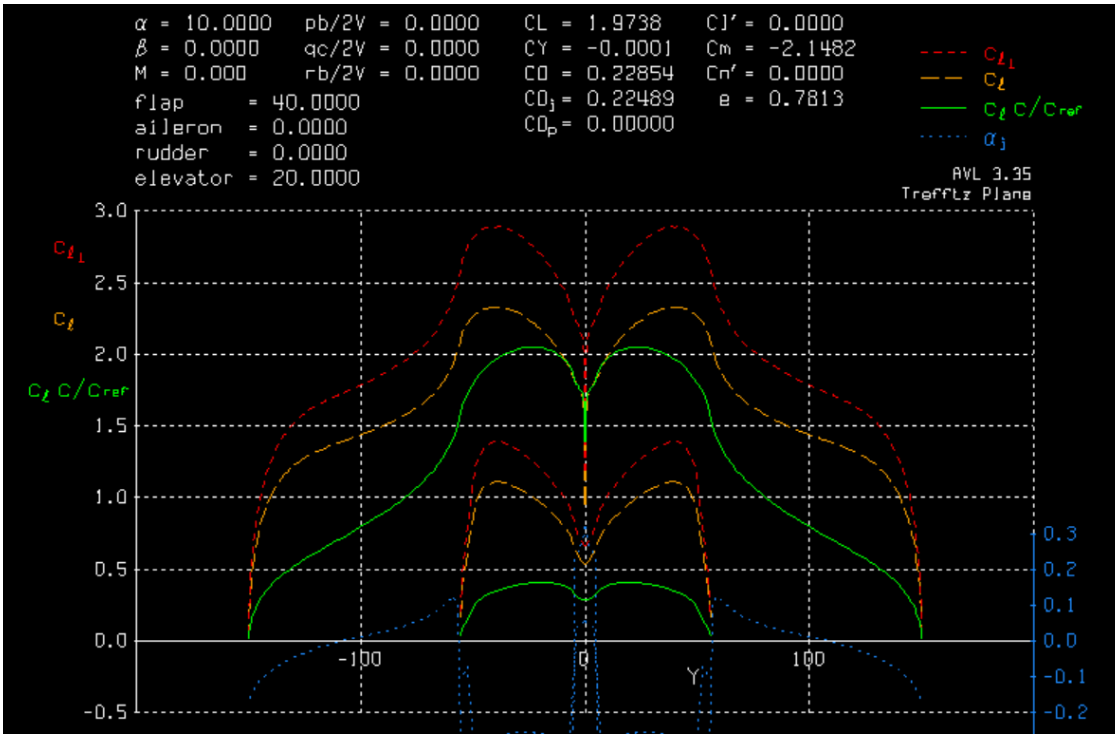


Figure 29. Lift Distribution During Landing

Structural Elements in the aircraft

The fuselage is strengthened by stringers and longerons along with bulkheads present at regular intervals.

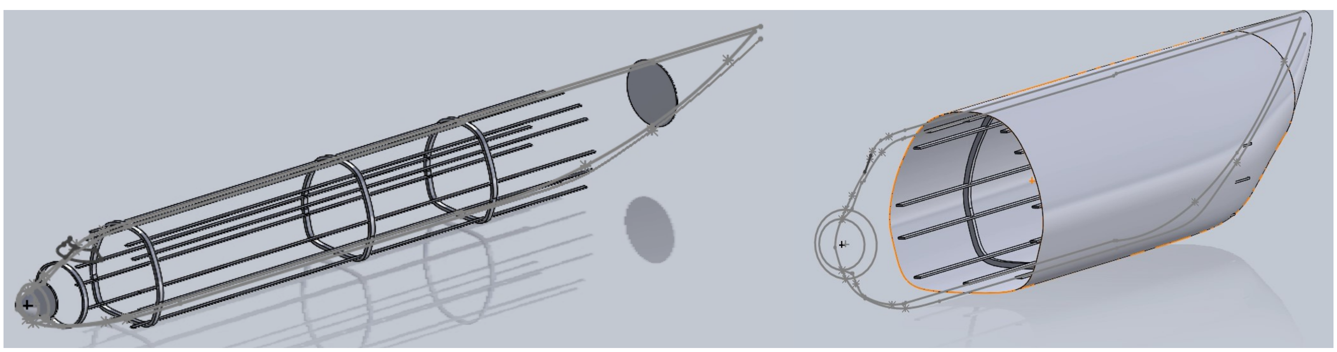
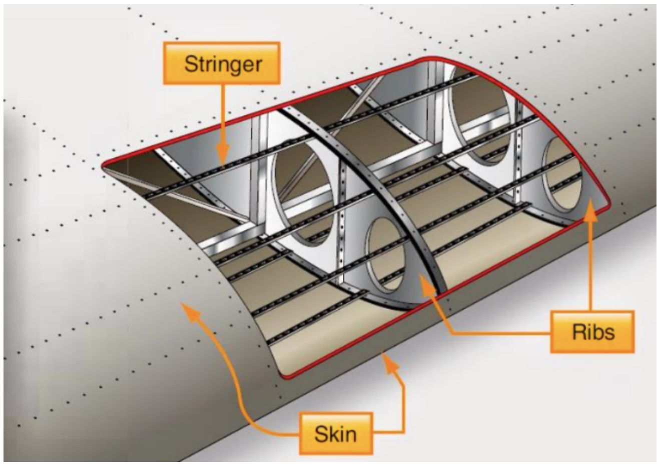


Figure 30. Fuselage Structural Elements

The wings are strengthened by stringers, ribs and spars, as shown in the figure below.



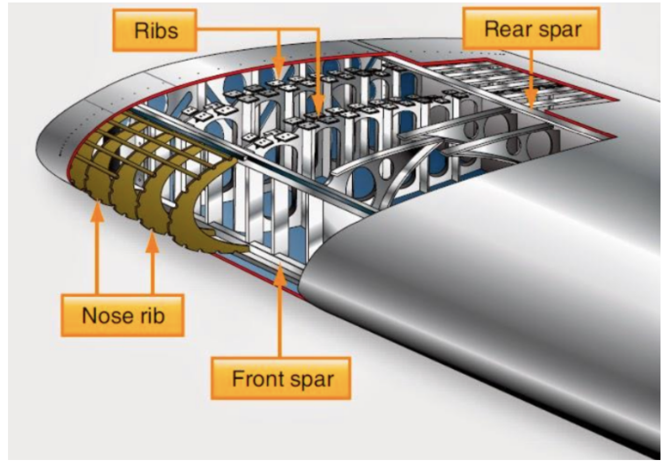


Figure 31. Wing Structural Elements, Source: AE481 Lecture Notes

As previously mentioned, the wings join the fuselage using a Spar Beam (also known as bending beam) that will pass through the top bay of the cargo bay.

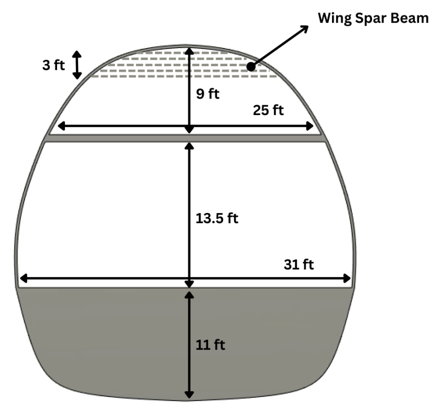


Figure 32. Structural Element Joining the Wings to the Fuselage and Location Within Fuselage

V-n Diagrams

The V-n diagrams for the aircraft are developed to determine the operational limits of the aircraft. The aircraft is designed to withstand load factors of +2.5 to -1 which is a common operating limit for many wide body cargo aircraft. This section details the gust envelope calculations for minimum and maximum weight at 20,000 ft. A short description of the maneuvering envelope V-n diagrams can be found in the appendix.

Maneuver Loads for Maximum Weight at 20,000 ft

In order to determine the maneuver loads, the team must calculate the required equivalent airspeeds. Since the aircraft is designed to fly at an altitude of 40,000 ft at a Mach number of 0.82, the TAS was calculated to be 653 knots. V_C is found by converting this to EAS (by using Eq. 11.1 from the metabook, for 20,000 ft), and V_{MO} is assumed to be 1.06 times V_C. The V_D is assumed to be 1.07 times the V_{MO}. The values were obtained as:

V_C = 477.39 knots

V_{MO} = 506.03 knots

V_D = 541.45 knots

Assuming the values of CL_{max} as 2.7 and -1.24, air density at sea level as 0.00237 sl/ft^3, and using the wing loading of 143 lbs/ft^2, the stall curves for maneuvering envelope were obtained.

Using a positive limit load factor of 2.5 and a negative limit load factor of -1, the various speeds were calculated as:

V_A = 197.73 knots

V_S = 125.06 knots

The maneuver part is complete (and shown in black solid lines in the plots). The horizontal line at top is given by n = 2.5, bottom one is given by n = -1. The right side is bounded by the V_D. There is a linear relation between V_D at n = 0, and V_c at n = 1. Using a similar procedure, the V-n diagram for minimum weight (empty payload) condition was obtained, as well using the minimum Wing Loading value equal to 72.93 lbs/ft^2. This is obtained by multiplying 0.51 (Ratio of empty weight to MTOW) to the original wing loading of the aircraft.

Maximum Weight with Gust Lines

In a similar manner to the initial maneuver load calculations, the gust load calculations were derived using eq. 11.10 in the Metabook detailing how the maneuvering load factor *n*, relates to the equivalent gust velocity, U_e, the gust alleviation factor, K_g, and the lift-curve slope, CL_α. Additionally, the gust envelope is dependent on both the equivalent airspeed, V_{EAS}, derived in the previous section, and the wing loading, W/S, also described in the previous section. Using the equations found in slide 21 of the V-n Diagram slides from class, both the gust alleviation factor and μ may be derived with the parameters found in the table below, along with the gravitational constant *g* in imperial units (32.17 ft/s^2). Finally, to obtain the equivalent gust velocities for the gust line calculation, figure 11.5 from the Metabook was used specifying the gust velocities as a function of altitude for the various design speeds. By working across the table at an altitude of 20,000 ft (specified in assignment deliverables), one can obtain the equivalent gust velocities used for the calculations for the three design speeds: V_B, V_C, and V_D; all of which are shown below.

Ue_{VB} = 66 ft/s

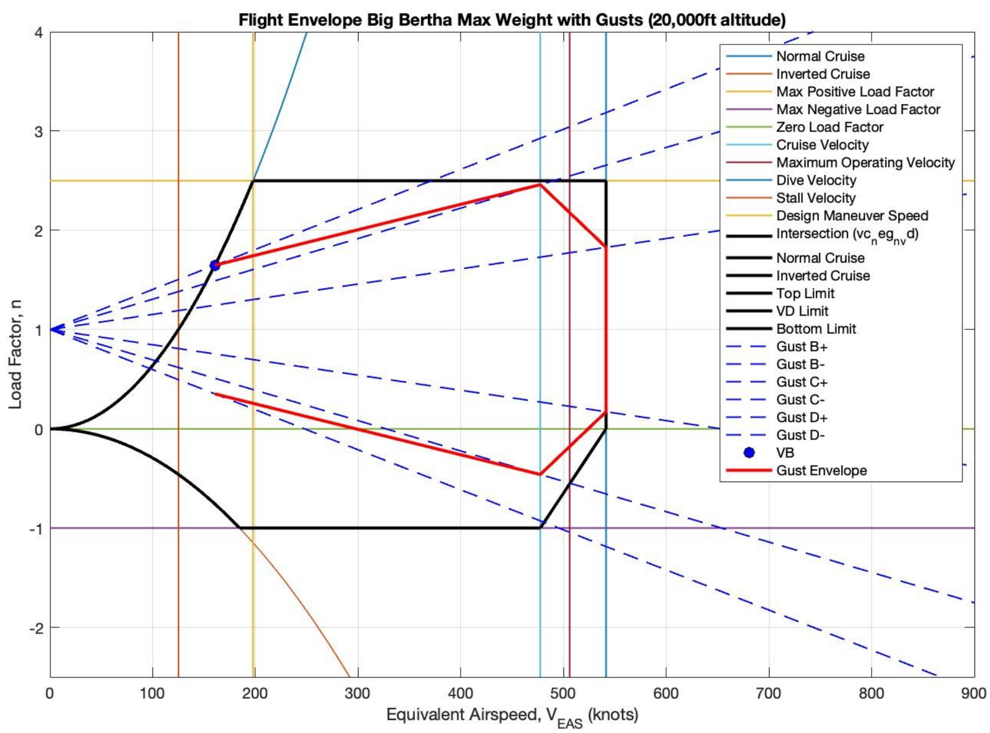
Ue_{VC} = 50 ft/s

Ue_{VD} = 25 ft/s

Table 17. Key Parameters Used For Gust Line Calculations

Parameter	Value
W/S	143 lbs/ft^2
$\bar{c}$	40.5 ft
$\rho_{20,000\ ft}$	0.001267 $slugs/ft^3$
CL_{α}	5.6707

Using these gust speeds, V_B = 160.55 knots, at the intersection of the stall line and the endpoints of gust lines for V_C and V_D were found at the points of intersection with the vertical lines at V = V_C and V_D. For the maximum weight condition, it was observed that the gust envelope is contained within the maneuver envelope. However, for the minimum weight condition, it was observed that the gust envelope exceeds the maneuver envelope.



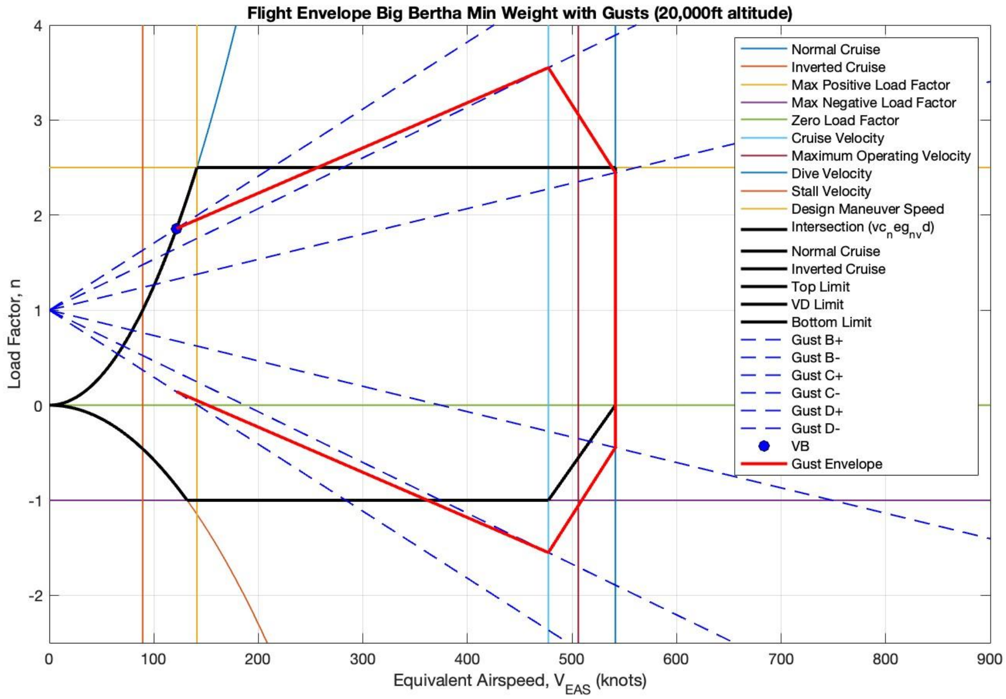


Figure 33. V-n Diagrams for Max (top) and Min (bottom) Weights with Gust Envelope

CL_{max} Values Used for the Stall Curves

The CL_{max} values for the V-n diagram calculations were obtained through AVL. These AVL calculations were done with the old wing sizing, before the design trade optimization in assignment 12. Specifically, with the AVL build file utilizing a wing span of 290 ft, with a root chord of 53.5 ft. Similar to assignment 11, the CL_{max} in the positive domain was determined with a flap deflection of 40º and an elevator deflection of 20º (landing configuration). It is important to note that this CL_{max} value was derived including high-lift devices, unlike the CL_{max} derived for the Boeing 777-200ER in the Metabook as this would result in worst-case loading on the wing. Finally, the CL_{min} value was derived for the aircraft through AVL assuming a clean configuration. As the aircraft class for the design project is a heavy military cargo aircraft, it is sufficient to assume the aircraft will not be able to perform any maneuvers upside-down, resulting in the clean configuration assumption.

Positive CL_{Max} = 2.7

Negative CL_{Max} = -1.24

Cost (Andrew)

The aircraft cost is separated into two parts, the cost of aircraft, and the aircraft direct operating cost. The cost of aircraft calculation was described in the *Objective Function* section, the Aircraft Metabook describes the method on calculating the direct operating cost (DOC). With the input parameters given, the cost breakdown for each component is shown in the table below.

Table 18. Finalized Design Input Parameters for Cost Estimation

Maximum Takeoff Weight	Maximum Takeoff Thrust	Fuel Weight	Airframe Weight	Block Time	Range
1,771,279 lbs	79,960 lbf	421,722 lbs	818,853 lbs	5 hr	2,500 nmi

Table 19. Finalized Design Production Cost; The engine price was the actual price as researched for PW-4000-112, described in the *Propulsion System* section; The calculation of the airframe cost is based on the maximum takeoff weight of the aircraft, as discussed in the *Objective Function* section.

Estimated Aircraft Cost (USD)	
Engines	$96,000,000 (Actual)
Airframe	$312,359,541 (Estimated)
Aircraft (total)	**$408,359,541**

Table 20. Estimated DOC of Final Design; The calculation of the cost depends on the maximum takeoff weight, maximum takeoff thrust, aircraft fuel weight, airframe weight, block time, and range.

Estimated Direct Operating Cost [USD]	
Crew	$12,845
Fuel	$332,903
Oil	$1,927
Airport	$4,856
Navigation	$4,064
Maintenance (airframe)	$23,069
Maintenance (engine)	$4,749
Cash Operating Cost (COC)	**$384,412**
Insurance	$9,330
Finance	$27,562
Depreciation	$16,795
Registration	$8,198
Fixed Operating Cost (FOC)	**$61,885**
Direct Operating Cost (DOC)	**$446,297**

Environmental Impact (Rawan, Andrew)

CO2 Emissions

The ICAO Scoping Report on Environmental Metrics of Relevance To the Global Aviation System was utilized to estimate the CO2 emissions for Lifting Large design missions, further analysis and comparison are shown in the appendix section *Environmental Impact Analysis*. The table below was employed to determine total CO2 emissions by Lifting Large in different configurations. The equation in the Formula/algorithm,

$$CO2 = fuel\, usage * 3.16$$

section was used to calculate the total CO2 emitted (in kilograms then converted to pounds) for three different configurations:

Table 21. Estimated CO2 emissions in pounds-CO2 for three different payload configurations and the desired mission range

Payload (lbs)	Range (nmi)	Fuel Weight (lbs)	CO2 Emissions (lbs*CO2)
430,000 (Full)	2,500	435,348	**1,375,700**
295,000	5,000	701,930	**2,218,099**
No Payload	8,000 (Ferry Flight)	944,903	**2,985,893**

CO2 Emissions per Pound of Payload

In order to calculate CO2 Emissions per pound of payload, the total emissions values from the previous section were used in the equation below:

$$CO2\ Emissions\ per\ Pound\ of\ Payload = Total\ CO2\ Emissions\ /\ Payload\ Weight$$

Table 22. The table shows Lifting Large's estimated CO2 emissions in the unit of pounds-CO2 per pounds for three different payload configurations and the desired mission range

Payload (lbs)	Range (nmi)	Fuel Weight (lbs)	CO2 Emissions per Pound of Payload (lbs*CO2/lbs)
430,000 (Full)	2,500	435,348	**3.1993**
295,000	5,000	701,930	**7.5190**
No Payload	8,000 (Ferry Flight)	944,903	**0**

* Note: No Payload configuration, despite being 0, is considered for the sake of comparison

CO2 Emissions per Pound Mile

In order to calculate CO2 Emissions per pound mile, the equation below was employed.

$$CO2\ Emissions\ per\ Pound\ Mile\ =\ Total\ CO2\ Emissions\ /\ Payload\ Weight\ /\ Range$$

Table 23. The table shows Lifting Large's estimated CO2 emissions in the unit of pounds-CO2 per pounds-nmi for three different payload configurations and the desired mission range

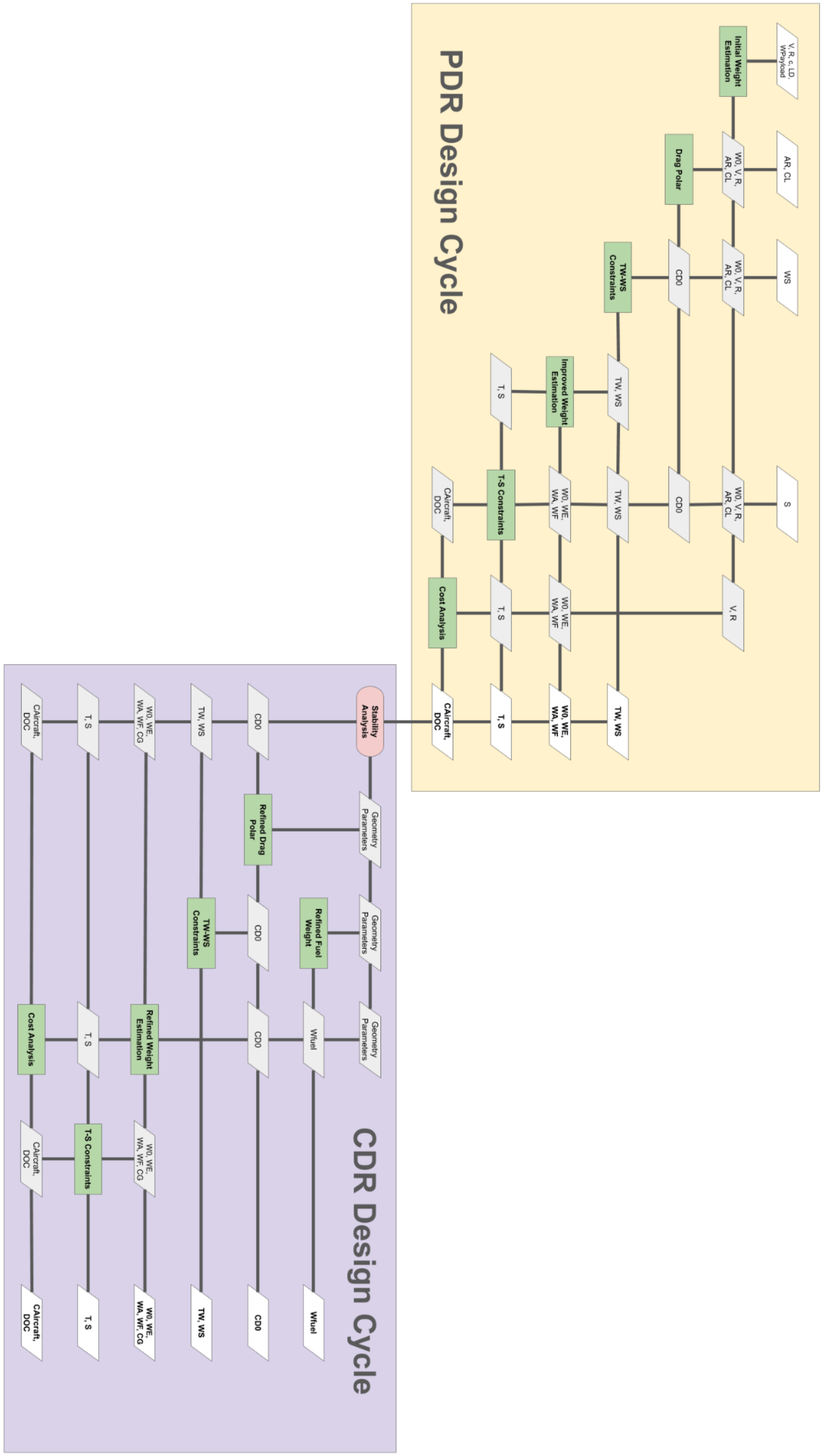
Payload (lbs)	Range (nmi)	Fuel Weight (lbs)	CO2 Emissions per Pound Mile [(lbs*CO2)/(lbs*nmi)]
430,000 (Full)	2,500	435,348	**0.0012797**
295,000	5,000	701,930	**0.0015038**
No Payload	8,000 (Ferry Flight)	944,903	**0**

* Note: No Payload configuration, despite being 0, is considered for the sake of comparison

NOx and H2O Emissions

The NOx calculations were derived mainly from the research paper: Simplified methodology of NOx and CO emissions estimation. A main takeaway from this paper was that "the higher the combustion temperature - the higher the efficiency of the engine and its performance, but greater toxicity of exhaust gases." Since Lifting Large employs highly powerful, efficient engines, this takeaway is highly applicable. Lifting Large anticipates NOx emissions of about 40 lbs per 2200 lbs of fuel used (about 0.018 NOx emissions per pound of fuel used). Based on historical analysis of aircraft H2O emissions and ICAO reports, the H2O emissions approximated for Lifting Large are higher than the NOx emissions at around 0.05 lbs of H2O emissions per pound of fuel used.

Computational Procedure (Andrew)



With the initial design defined in the preliminary phase, the basic parameters such as wetted area from CAD, engine specifications from engine choice, and the lifting surfaces data from aerodynamics were available for a more in-depth analysis. During the critical design phase, the team took the approach of component-breakdown analysis to refine the design parameters further.

Drag Polar

First, a more detailed analysis was conducted on the drag polar. With the available aerodynamics and CAD data, the analysis takes into account the component-wise parasitic drag, including individual drags for wing, vertical stabilizer, horizontal stabilizer, fuselage, and nacelles. Other miscellaneous drags such as fuselage upsweep drag due to separation, flap drags, landing gear drag, and trim drags were also considered for each different flight configurations. The improved drag code also outputs the drag polar plot for all flight configurations (takeoff, retracted gear; takeoff, deployed gear; landing, retracted gear; landing, deployed gear; cruise, trimmed drag; clean, no drag). The calculated parasite drags for each flight configuration are then being fed into the refined weight estimation.

Fuel Weight

Second, an improved fuel weight estimation was also being considered. To improve the fuel weight ratio estimation, the climb and cruise stages are being broken down into multiple segments. As referred in the Lecture 13 Sizing Refinement, each segment updates the weight and the drag it experienced, which are then fed into the Breguet Range Equation to calculate the fuel weight for the specific segments. This method allows a more accurate capture of the fuel consumption being used during the flight. For the other flight stages' fuel weight ratio, including warmup/taxi, takeoff, landing, are estimated from the historical statistics. By the end of the calculation, the code scaled the fuel weight ratio by 1.06 to account for the trapped fuel. Altogether, the code outputs the improved fuel weight ratio to be used by the refined weight estimation code for the fuel weight calculation.

$$\frac{W_{i+1}}{W_i} = exp\left(-\frac{C*\Delta h_e}{V(1-D/T)}\right) \text{ // for climb}$$
$$\frac{W_{i+1}}{W_i} = exp\left(-\frac{R*C}{V(L/D)}\right) \text{ // for cruise}$$

Weight and CG

With improved calculation of drag and fuel weight, the refined weight estimation also takes the approach of component-breakdown method. By first separating the aircraft into parts: engine, lifting surfaces (wing, horizontal stabilizer, vertical stabilizer), fuselage, landing gear. The component weights were estimated using the approximate empty weight buildup method from Raymer, through the correlations between the wetted area of the components and the weights. Additionally, the weight of the wing was calculated using the Raymer's regression for cargo/transport aircraft, using the equation:

$$W_{wing} = 0.0051(W_{dg}N_z)^{0.557}S_w^{0.649}AR^{0.5}(t/c)_{root}^{-0.4}(1+\lambda)^{0.1}(\cos\Lambda)^{-1}S_{csw}^{0.1}$$

Other weight components such as fuel, payload, and crew weights were also computed in accordance with the mission requirements, while the fuel weight was produced from the improved weight estimation code. Lastly, the refined weight estimation also approximated the weight of all other components that were not considered in the code to be around 17% of the maximum takeoff weight. For such, the code loops the weight calculation until it converges, resulting in the final refined weight estimation.

Furthermore, the refined weight estimation code also outputs the CG location for each component. Again, by using the approximate empty weight buildup method from Raymer, the CG location for each weight component was then being calculated given the characteristic geometries of the component. The CG location of the entire aircraft was then calculated using the component CGs, where the results are the aircraft CG location for MTOW configuration, aircraft CG location for takeoff configuration, aircraft CG location for pre-landing configuration, and the aircraft CG location for landing configuration. These CG locations are being used in the CG excursion plot for analysis on the aircraft's stability.

Design Plot

When the weight and the geometry of the aircraft were defined, the new design parameters were being plotted with the constraints from RFP. The plotting of the design point with the constraints provided valuable insight into the further refinement of the aircraft design.

First, the design point of the current aircraft can be calculated: calculate the wing loading (W/S) by dividing the wing reference area from the maximum takeoff weight; calculate the thrust-to-weight ratio (T/W) by dividing the maximum takeoff weight from the maximum takeoff thrust. The design point is then being plotted together with the constraints in the T/W - W/S plot. This design plot contains information on if the current design point is feasible, and how to further refine the design.

In addition, the design parameters of maximum takeoff thrust and the wing reference area of the current aircraft is also plotted in the T - S plot together with the constraints. Same as T/W - W/S plot, this design plot contains information on the feasibility of the current parameters, and how to further optimize the design in accordance with the cost objective function contour.

By analyzing the T/W - W/S and T - S plots, a more refined combination of maximum takeoff thrust and wing reference area can then be updated. With the updated thrust and reference area, the design cycles back to refine the geometry in accordance with the stability analysis. The updated geometries and engine thrust will then again be analyzed in this design cycle, until the most optimal point is chosen.

Cost

The cost objective function is embedded in the T - S Plot, which serves as the additional parameter to optimize the design. The cost function takes into account the weight estimation of the aircraft, and estimates the unit production cost of the aircraft using the historical regression analysis of the commercial jet transport aircraft, with the equation below, the CEF indicates the cost escalation factor:

$$C_{aircraft} = 10^{(3.3191+0.8043\, log_{10}(MTOW))}(CEF)$$

This function is overlaid as a contour in the T - S plot, where the design point would be chosen depending on the trade between the cost and the performance.

Stability

Ultimately, as the above parameters were defined in a single design loop (drag, weight, CG, cost), the design was then passed through a stability analysis in the form of AVL to validate sizing parameters, where the new geometries were defined, such as wing area, fuselage size, tail configurations, etc. The new geometric parameters were then being passed into the design loop again for further analysis and refinement. The design loops until the change becomes minimal, thus reaching an optimal design point.

Conclusion (Rawan, Ian)

This CDR Report outlined the work done at the design stage where Lifting Large Design Co. ensured the design satisfied the deliverables requested in the RFP provided by the Department of Defense. Included was information on the finalized design decisions, cost estimations, explanations of the design process, and more. Some key parameters from the finalized design are shown below.

Table 24. Key Performance and Design Parameters of the Final Design

MTOW (lbs)	**Unit Cost ($)**	**Engine**	**CO2 Emissions per Pound of Payload at Max Payload (lbs*CO2/lbs)**	**Reference Area (ft^2)**	**Range (nmi)**	**Payload (lbs)**
1,771,279	408,359,541	Pratt & Whitney 4000-112	3.1993	12,000	2,500 - 8,000	Empty - 430,000

In addition to the key parameters related to the design above, the finalized design by Lifting Large was able to meet the deliverables requested in the RFP through a key few design choices. First, the commonly used tube and wing configuration led to significant structural weight prevention compared to a BWB or Braced Wing configuration. Second, the high wing and t-tail combination aided in aerodynamics and stability, while also allowing for minimal and efficient loading/unloading. Third, the six engine configuration led to a sharp reduction in unit production cost when compared to a larger, four engine configuration.

The conformance plan that follows the conclusion of the CDR stage includes next steps of designing the interior structural layout, including hydraulic mechanisms, a fuel dispersion system, cargo loading mechanisms, and the cockpit; along with further optimizing the cost and aerodynamics of the aircraft. Furthermore, Lifting Large needs to analyze the design for common failure modes and possible points of redundancy in the main aircraft life support systems. Additionally, parameters outside of the RFP will need to be considered including certification.

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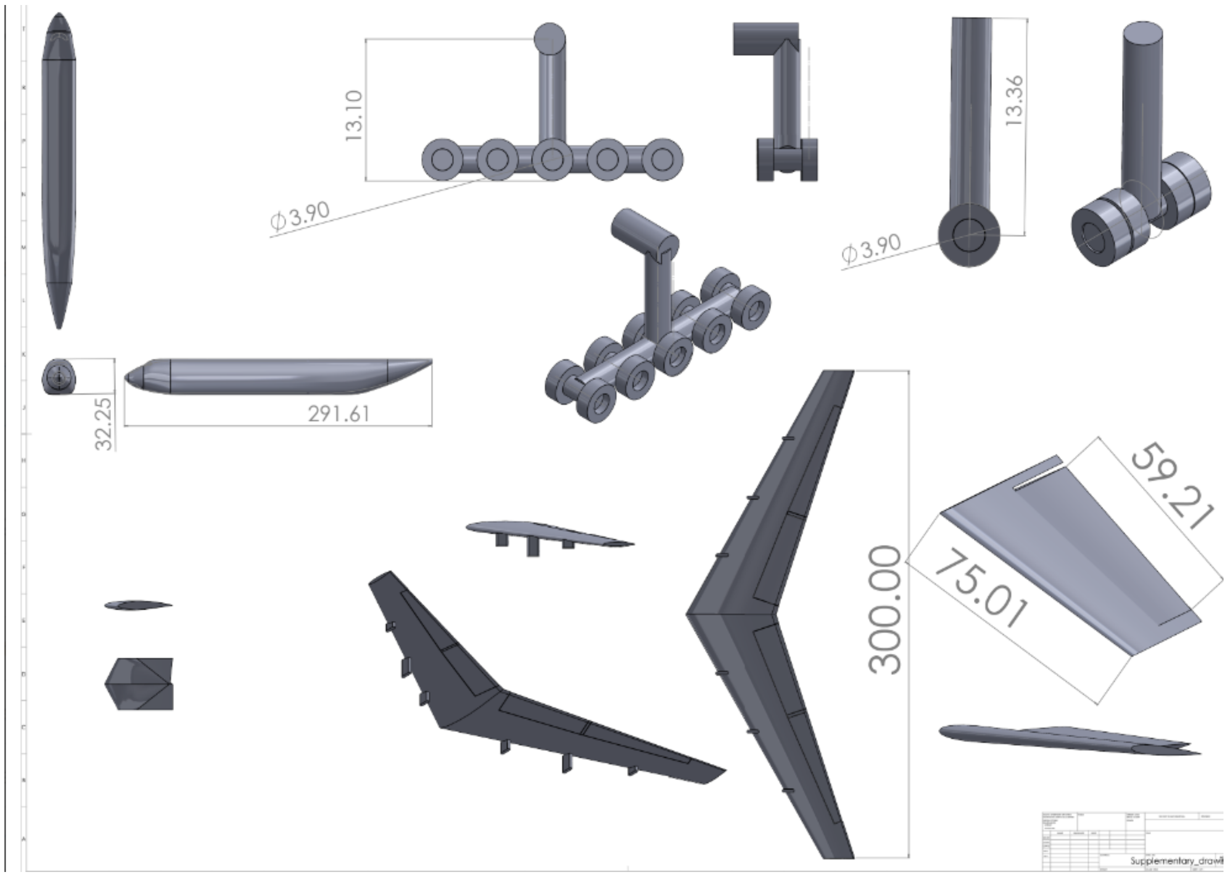
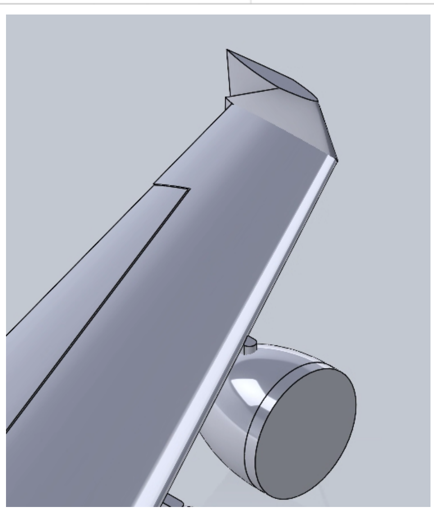
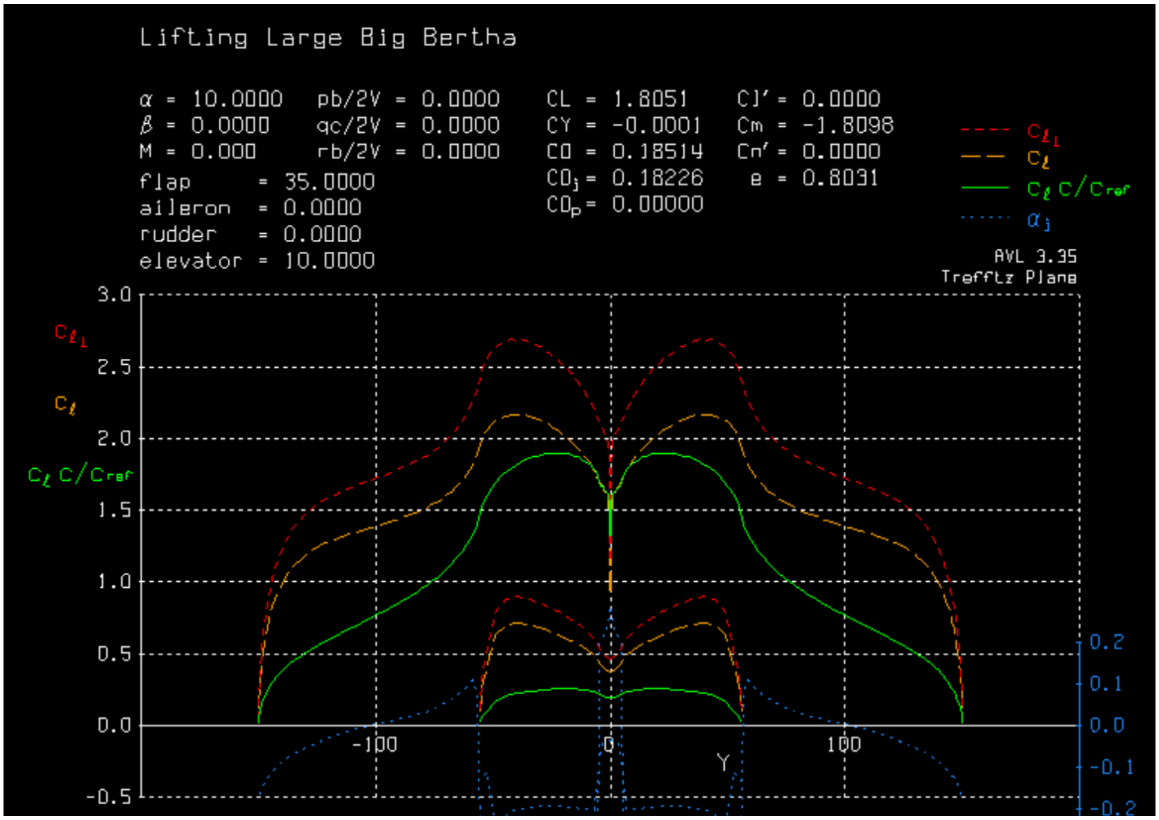
Appendix

Supplementary Engineering Drawings

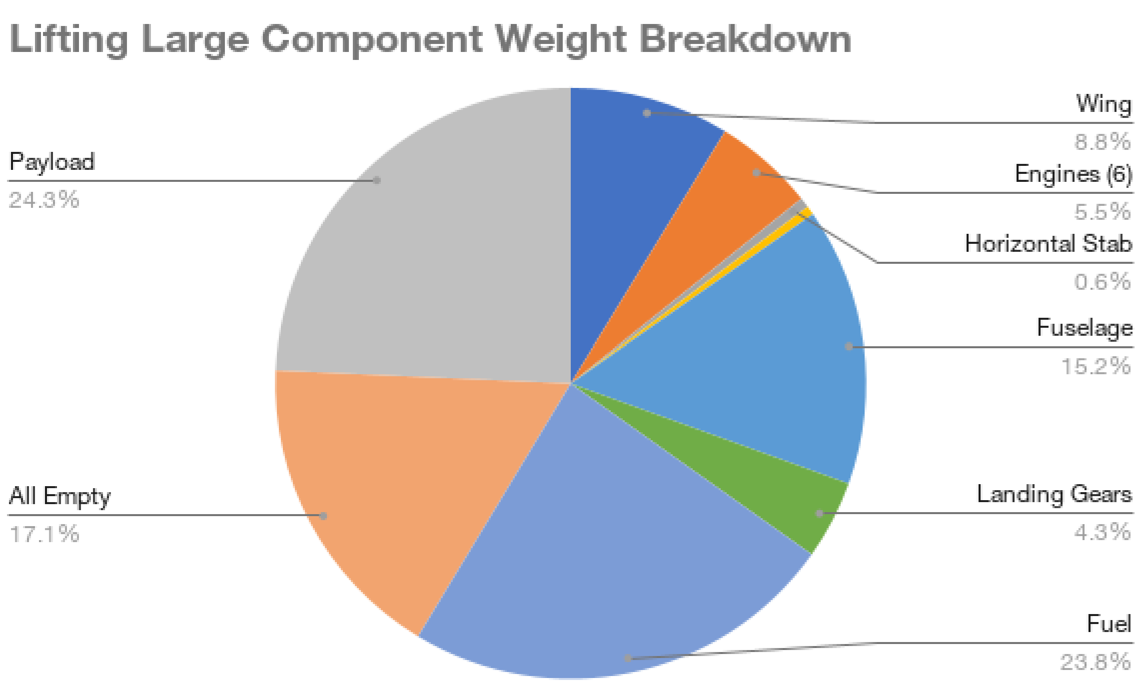




Spanwise lift distribution for takeoff



Visualization of relative component weights of the current design



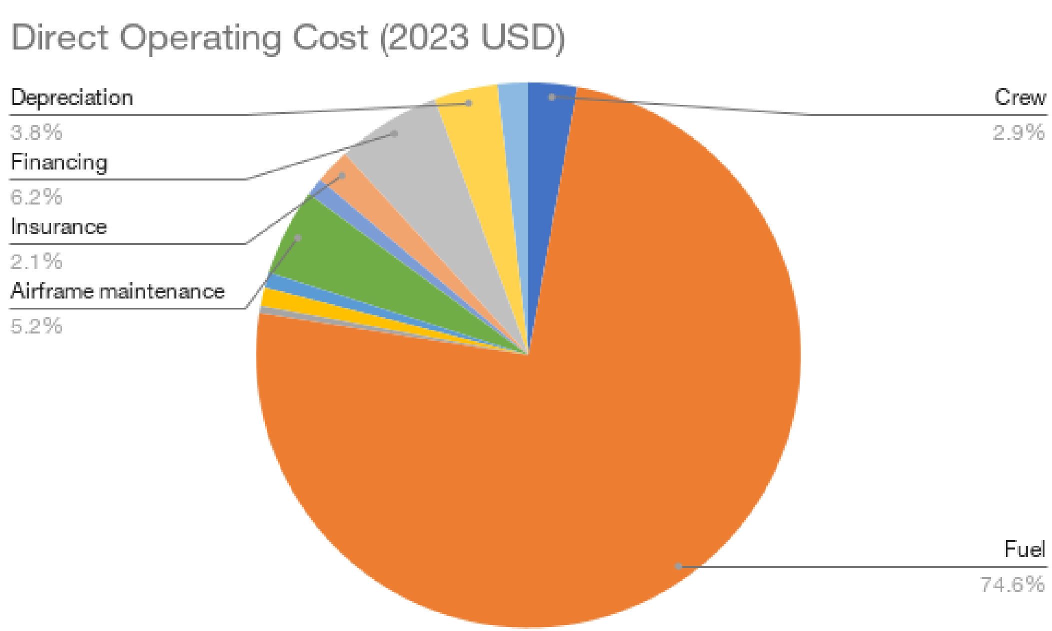
Cost Breakdown for USD per pounds-nmi

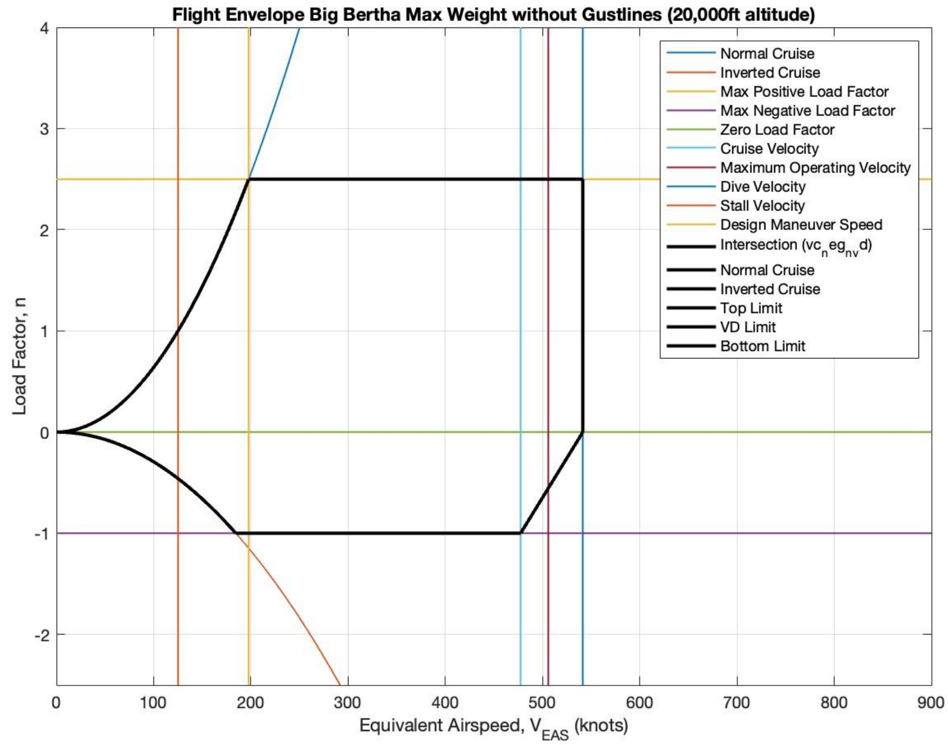
Table detailing Lifting Large's estimated direct operating cost in the unit of USD per pounds-nmi given different component's contribution at the max payload configuration (430,000 lbs payload, 2,500 nmi range).

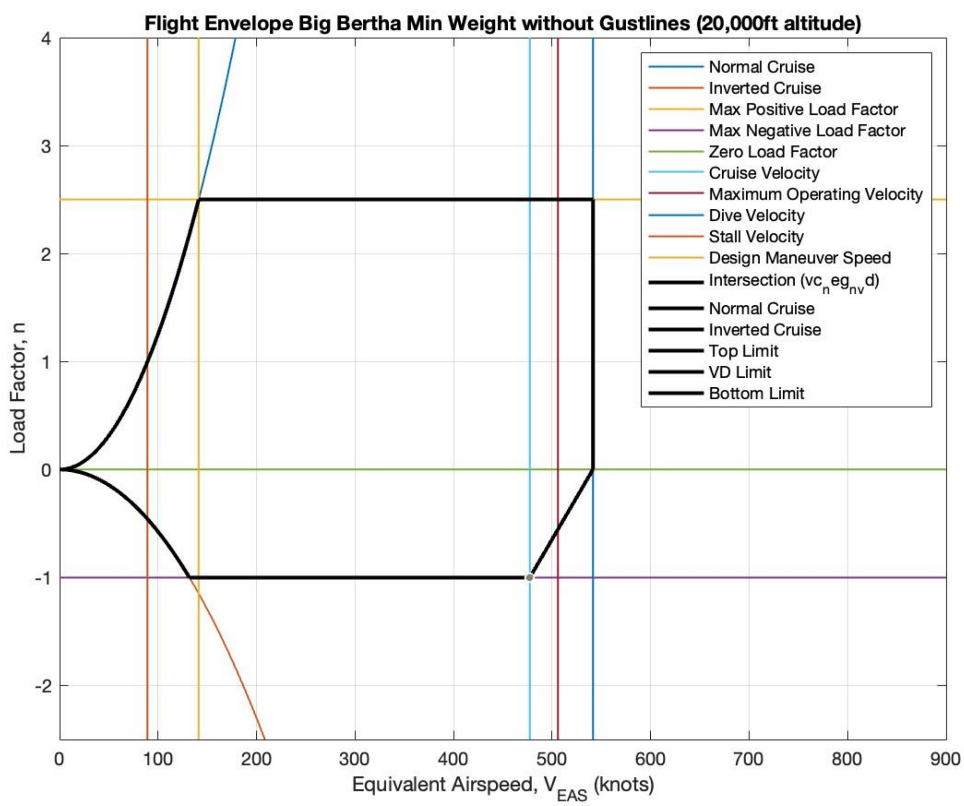
Estimated Direct Operating Cost [USD/lbm-nmi]

Crew	0.000002675847031945500
Fuel	0.000069352387852156300
Oil	0.000000401417594652554
Airport	0.000001011553113058310
Navigation	0.000000846672147763724
Maintenance (airframe)	0.000004805909432538910
Maintenance (engine)	0.000000989278448934986
Insurance	0.000001943781036733260
Finance	0.000005741879266044850
Depreciation	0.000003498805866119870
Registration	0.000001707869960708570
Direct Operating Cost (DOC)	0.0000929754017506568

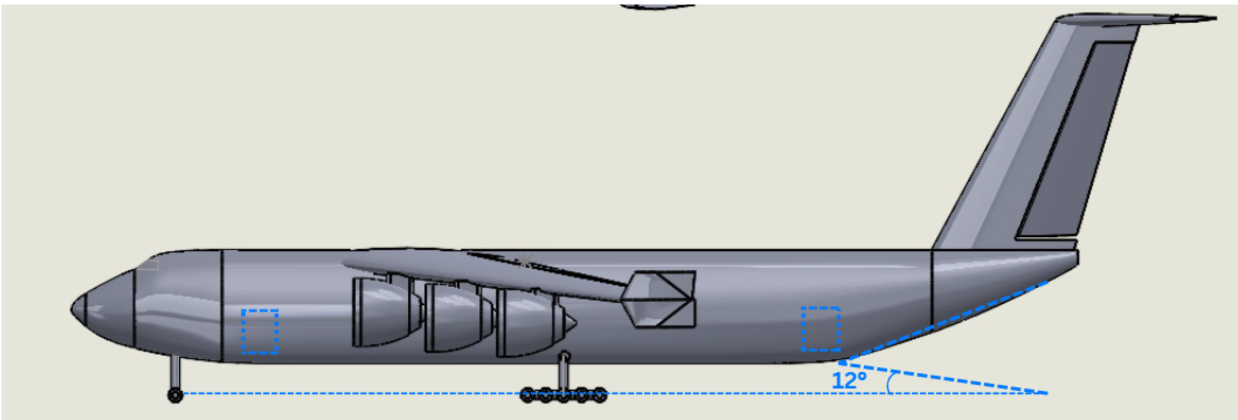
Visualization of the DOC contribution from each component in percentage. Note that the fuel cost is the major contributor to the Lifting Large's direct operating cost, ranging up to 74.6% of the DOC.







V-n Diagrams for Final Design at Max Weight (top) and Min Weight (bottom)



Rear Loading Ramp Configuration

<u>Environmental Impact Analysis</u>

CO2 Emissions per Passenger Mile

In order to calculate CO2 Emissions per passenger mile, each passenger was assumed weight of 190 lbs as stated by the FAA Standards, the equation below was employed.

$$CO2\ Emissions\ per\ Passenger\ Mile\ =\ Total\ CO2\ Emissions\ /\ Passenger\ Number\ /\ Range$$

Table. The table shows Lifting Large's estimated CO2 emissions in the unit of pounds-CO2 per passenger-nmi for three different payload configurations and the desired mission range

Payload (lbs)	Range (nmi)	Fuel Weight (lbs)	CO2 Emissions per Passenger Mile [(lbs*CO2)/(pax*nmi)]
430,000 (Full)	2,500	435,348	**0.243164**
295,000	5,000	701,930	**0.285653**
No Payload	8,000 (Ferry Flight)	944,903	**0**

* Note: No Payload configuration, despite being 0, is considered for the sake of comparison

CO2 Emission Comparison with Honda CR-V

To analyze the sustainability of Lifting Large's CO2 emissions, the CO2 emissions for a 2-seater 2024 Honda CR-V was being compared. To calculate the emissions of Honda CR-V, the fuel consumption of the vehicle was used. With a gas mileage of 40 gallons per mile, the fuel weight for each range was computed. The ICAO CO2 emissions equation was used to estimate the CO2 emissions of the Honda CR-V. Lastly, the same equation to calculate the CO2 emissions per passenger mile was again used, the calculated values are listed in the table below:

Table. The table shows a 2-seater 2024 Honda CR-V's estimated fuel consumption and the CO2 emissions in the unit of pounds-CO2 per passenger-nmi for three different range

Range (nmi)	Fuel Weight (lbs)	CO2 Emissions (lbs*CO2)	CO2 Emissions per Passenger Mile [(lbs*CO2)/(pax*nmi)]
2,500	445	1,407	**0.281450**
5,000	893	2,821	**0.282148**
8,000	1,426	4,507	**0.281713**

When comparing the values with Lifting Large's CO2 emissions per passenger mile, it is observed that the Lifting Large's CO2 emissions are roughly equivalent to the 2-seater 2024 Honda CR-V CO2 emissions. This drives the aircraft to be decently sustainable and environmentally friendly.